

전국 대학생 AI
자율주행 경진대회
교육 워크숍

Subject: 개발환경 구축

Automation Lab.



PC

■ PC 하드웨어 환경



운영체제: Window10

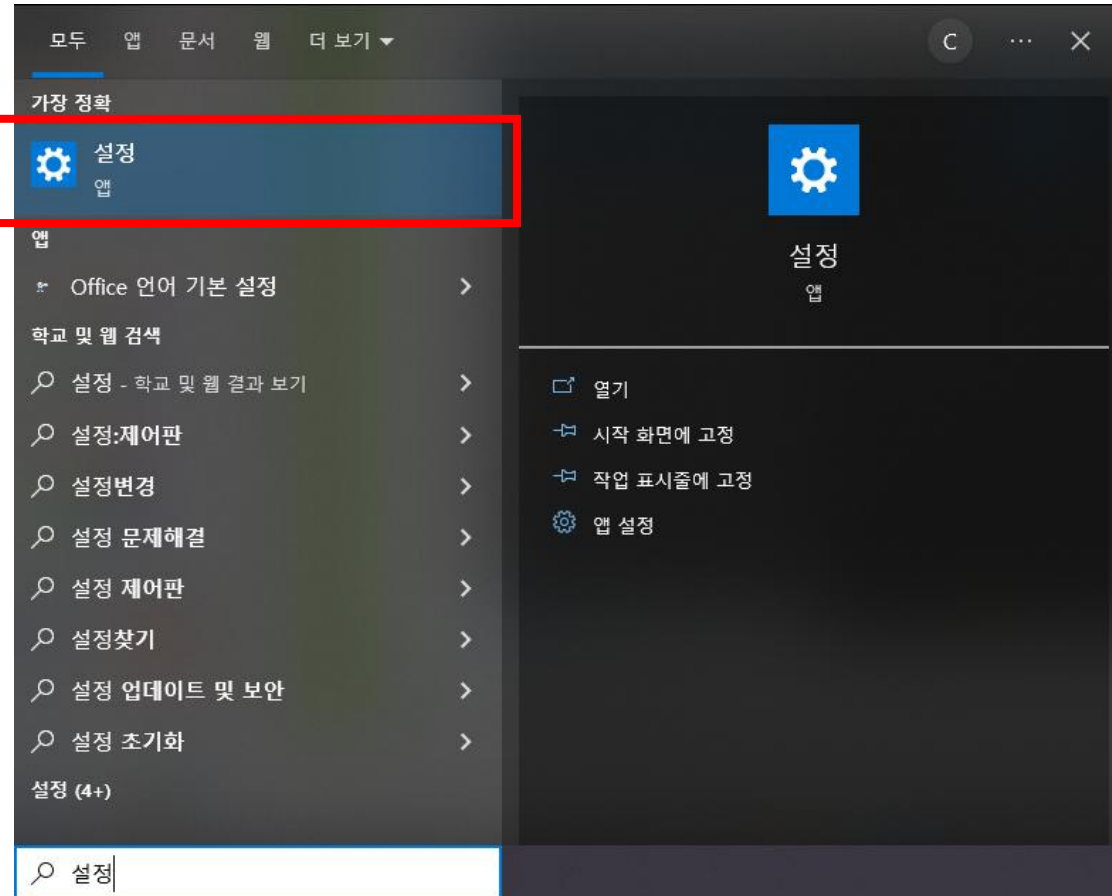


아키텍처: x86, x64

PC

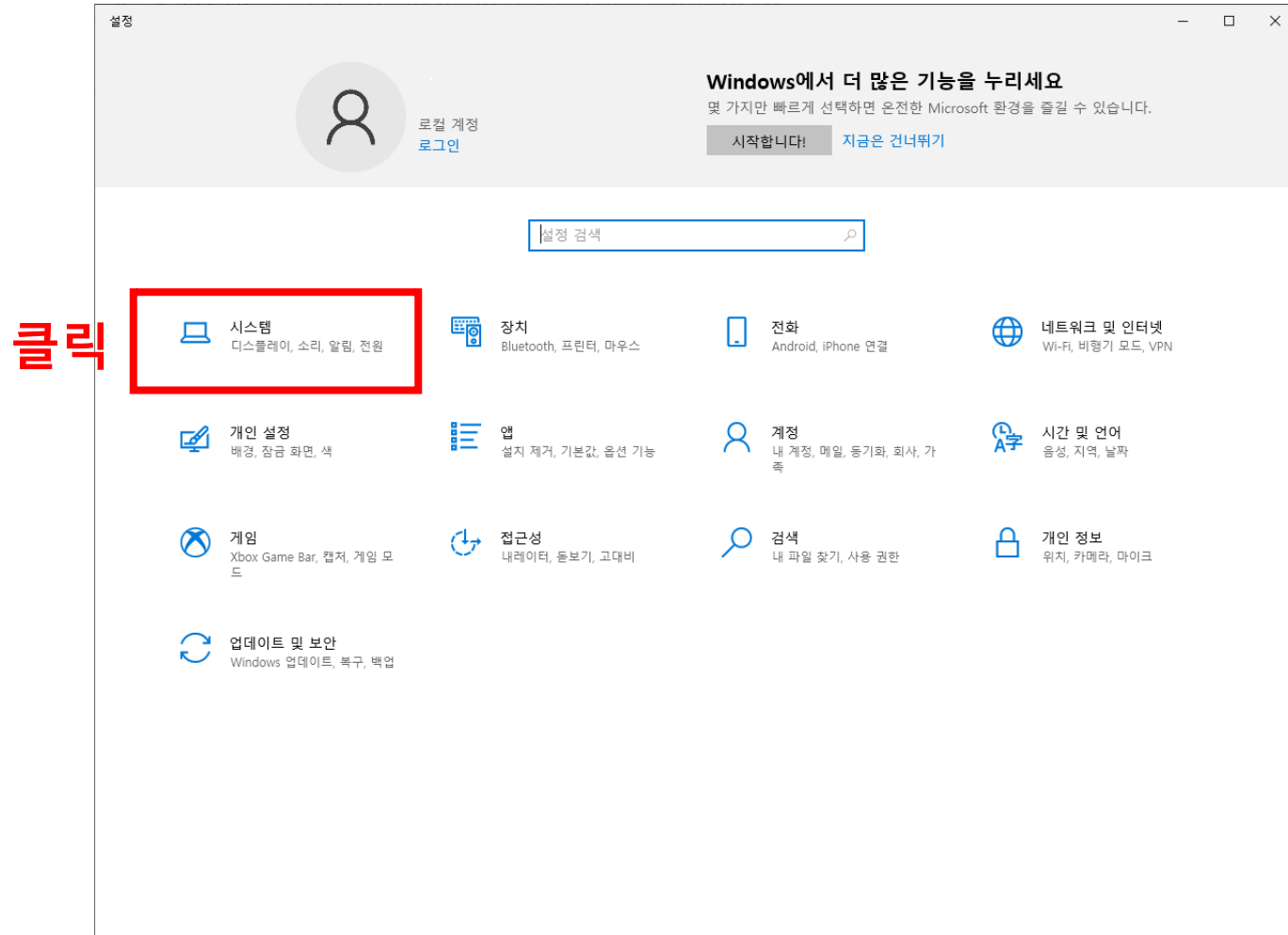
■ PC 하드웨어 사양 확인

클릭



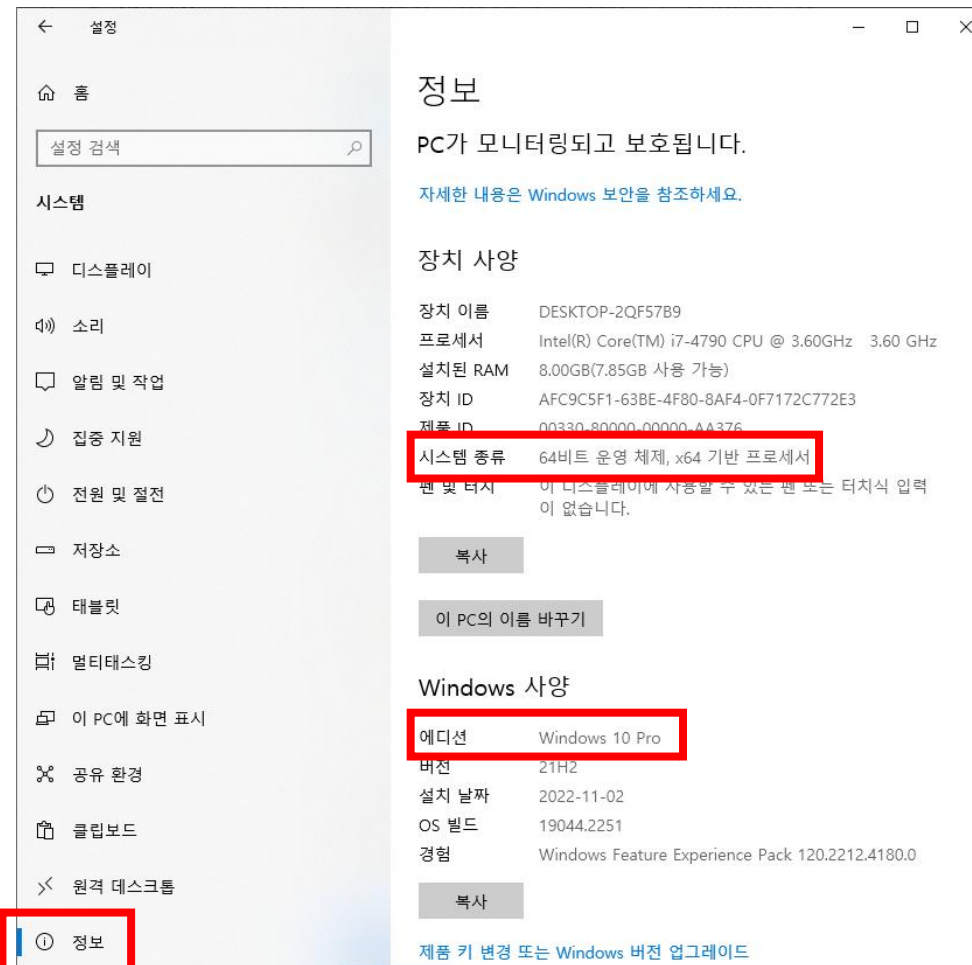
PC

■ PC 하드웨어 사양 확인



PC

■ PC 하드웨어 사양 확인

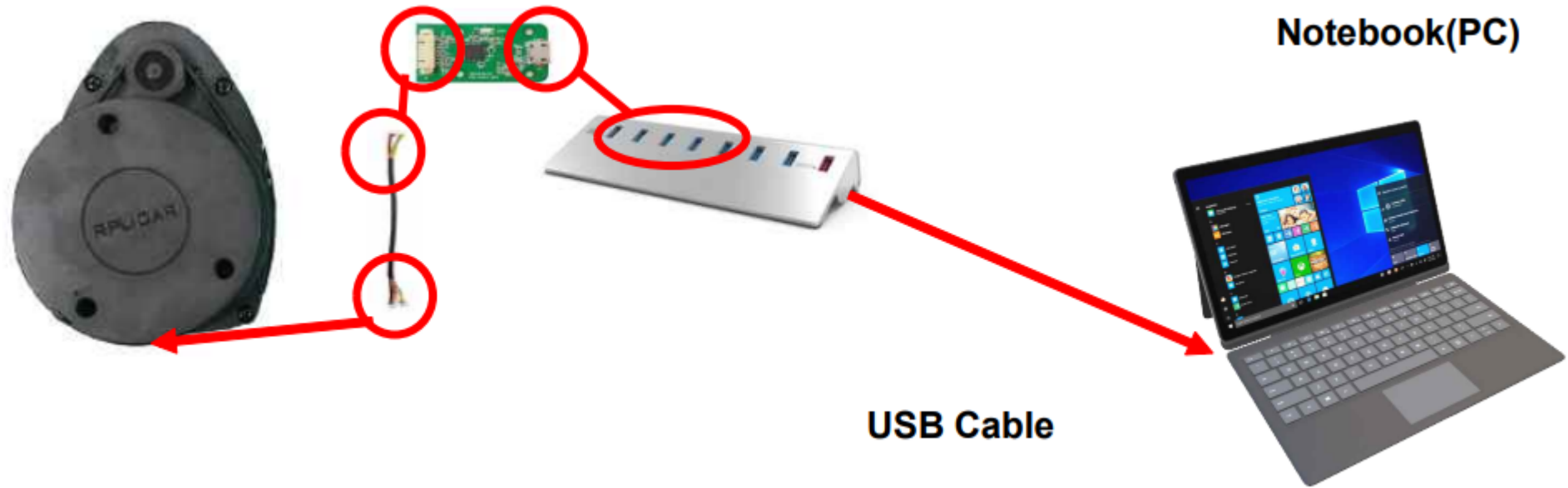


클릭

① 정보

LiDAR

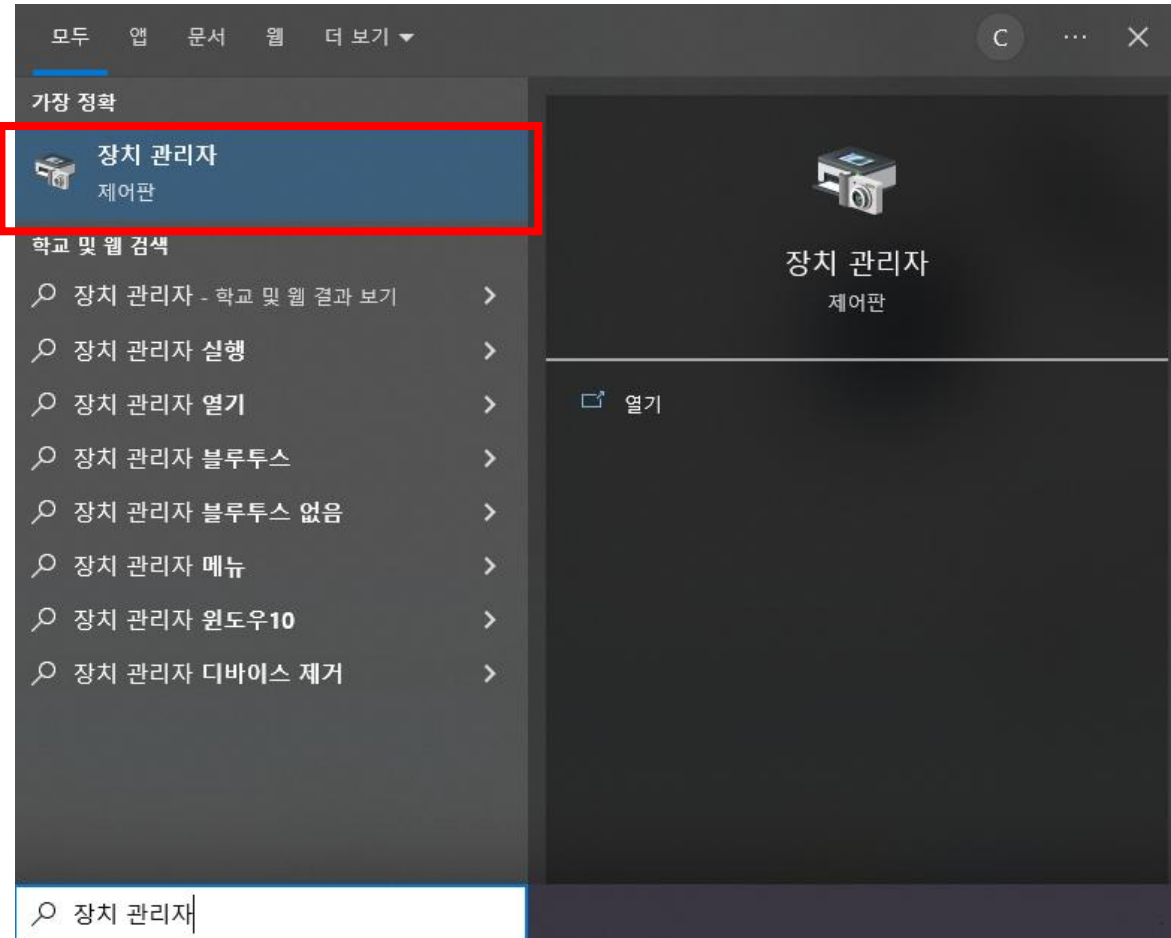
■ 라이다(LiDAR)



LiDAR

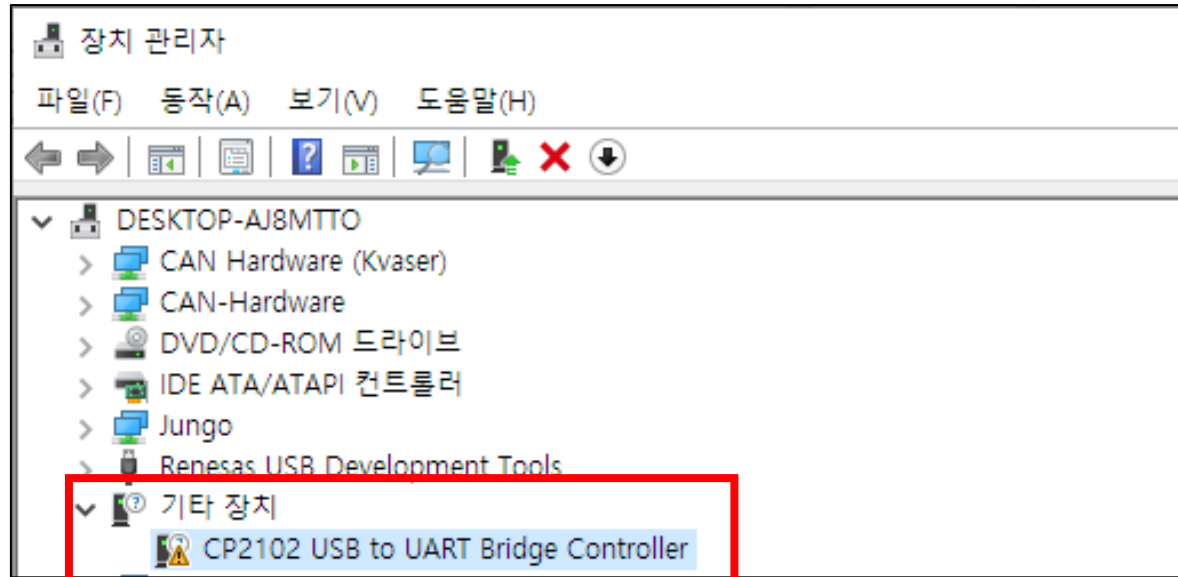
■ 라이다 장치 인식 확인

클릭



LiDAR

■ 라이다 장치 인식 확인



기타장치 CP2102 USB 인식 확인

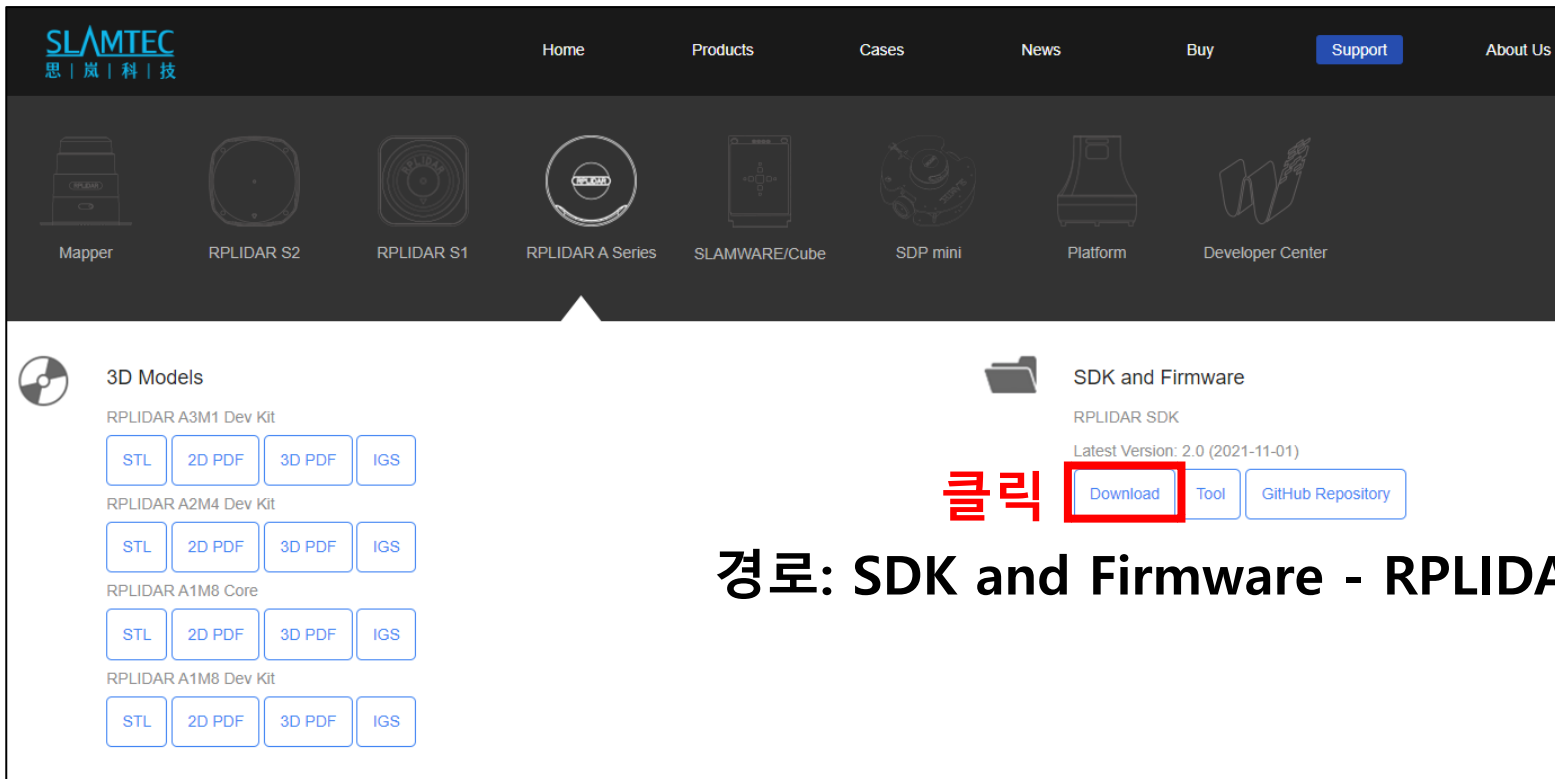
인식 안될 시, 라이다 USB 포트 연결 확인

LiDAR

■ 라이다 드라이버 설치

다운로드 페이지:

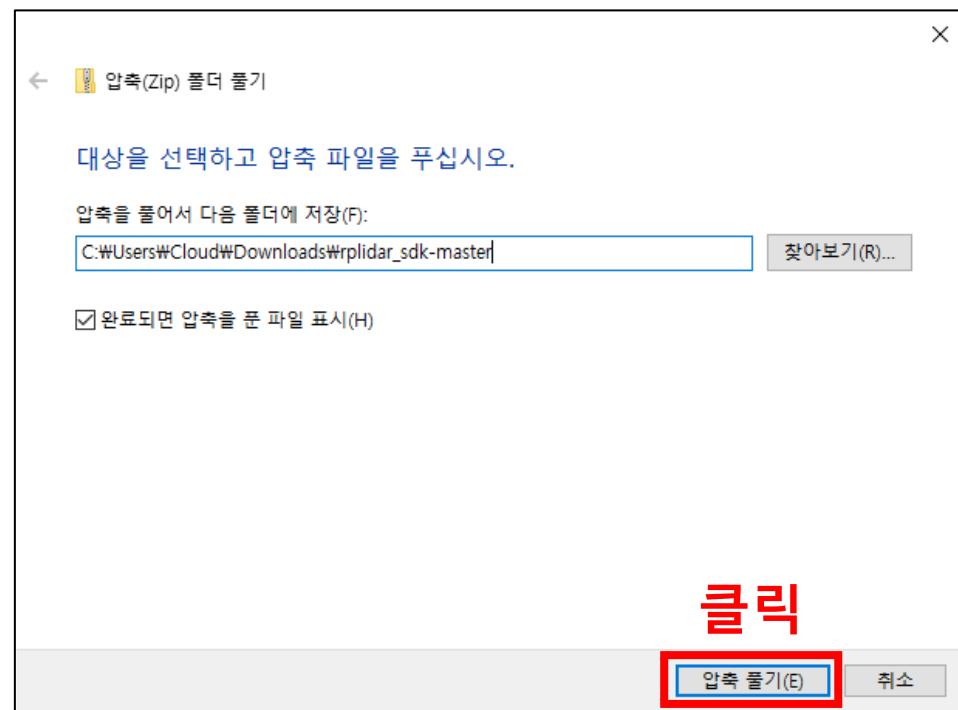
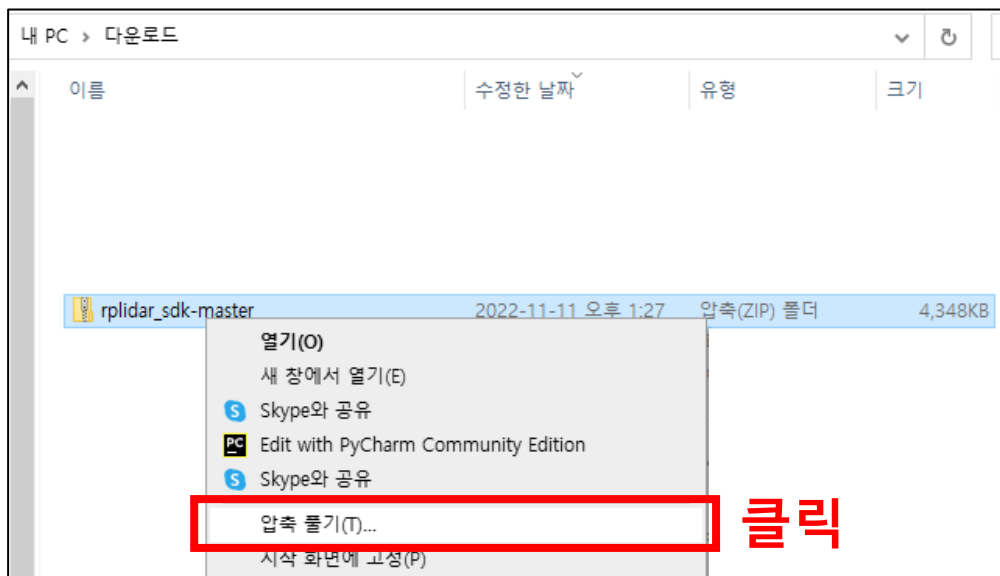
<https://www.slamtec.com/ko/support#rplidar-a-series>



경로: SDK and Firmware - RPLIDAR SDK - Download

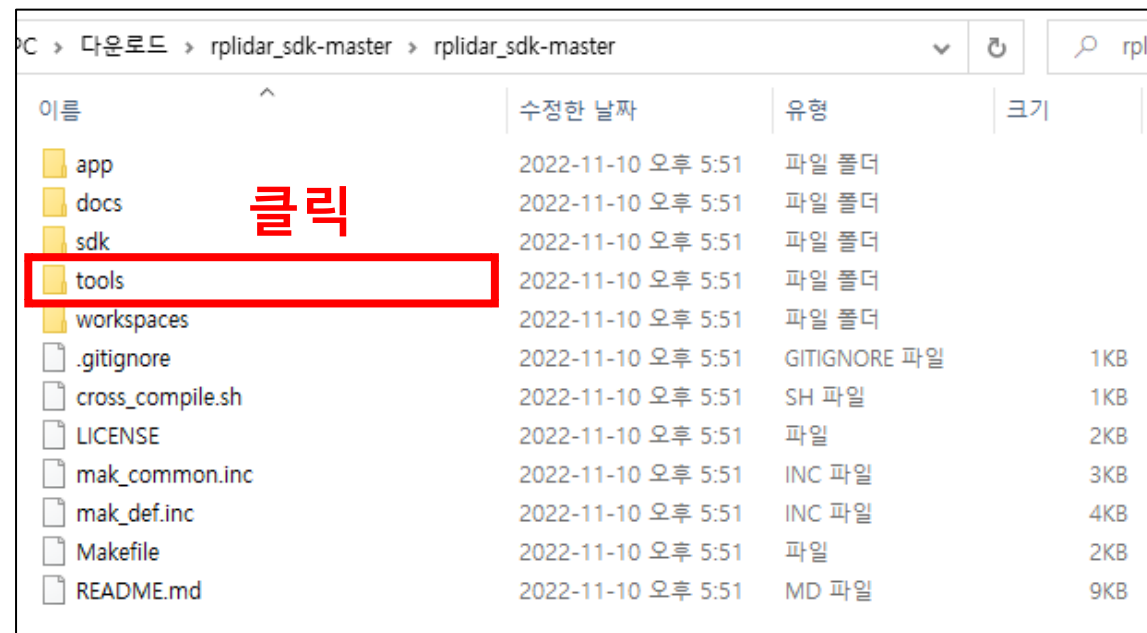
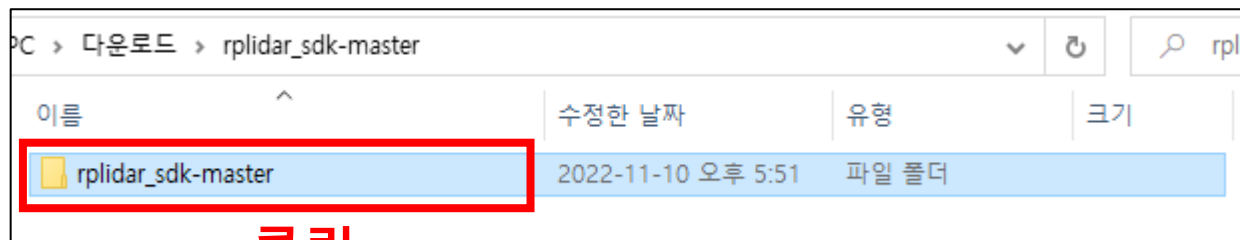
LiDAR

■ 라이다 드라이버 설치



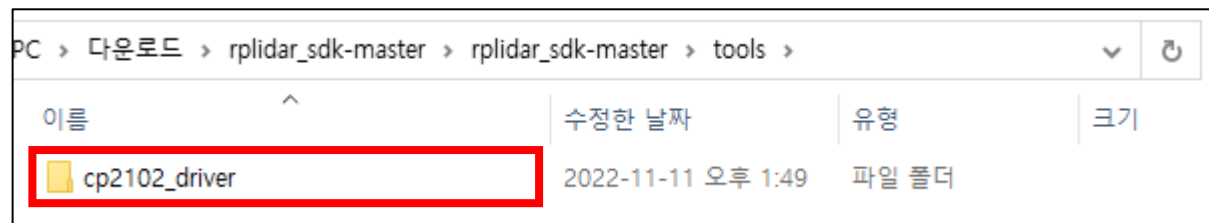
LiDAR

■ 라이다 드라이버 설치

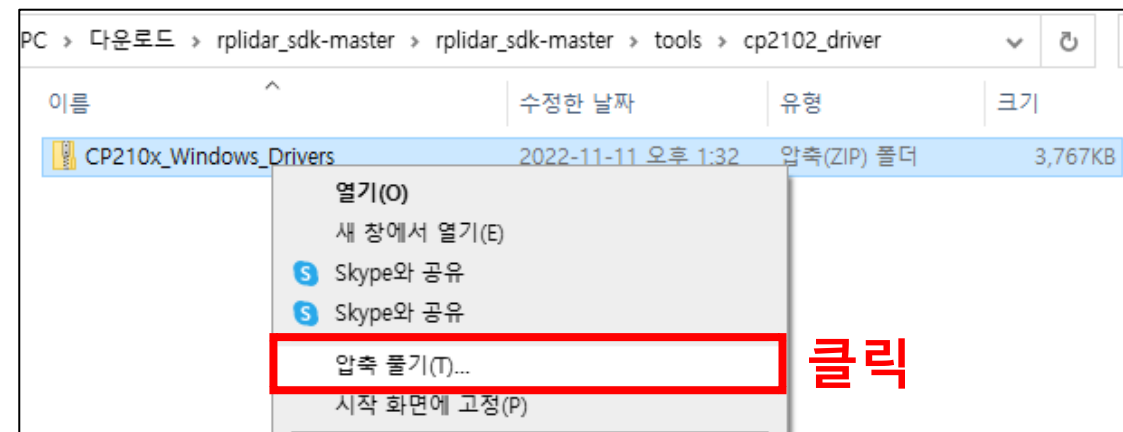


LiDAR

■ 라이다 드라이버 설치



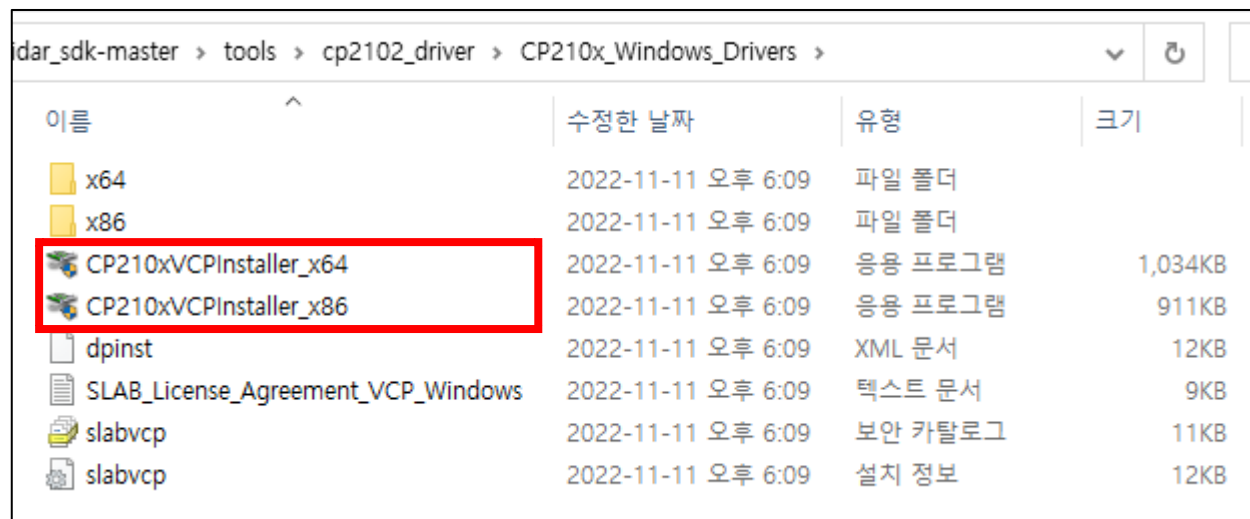
클릭



클릭

LiDAR

■ 라이다 드라이버 설치



이름	수정한 날짜	유형	크기
x64	2022-11-11 오후 6:09	파일 폴더	
x86	2022-11-11 오후 6:09	파일 폴더	
CP210xVCPInstaller_x64	2022-11-11 오후 6:09	응용 프로그램	1,034KB
CP210xVCPInstaller_x86	2022-11-11 오후 6:09	응용 프로그램	911KB
dpinst	2022-11-11 오후 6:09	XML 문서	12KB
SLAB_License_Agreement_VCP_Windows	2022-11-11 오후 6:09	텍스트 문서	9KB
slabvcp	2022-11-11 오후 6:09	보안 카탈로그	11KB
slabvcp	2022-11-11 오후 6:09	설치 정보	12KB

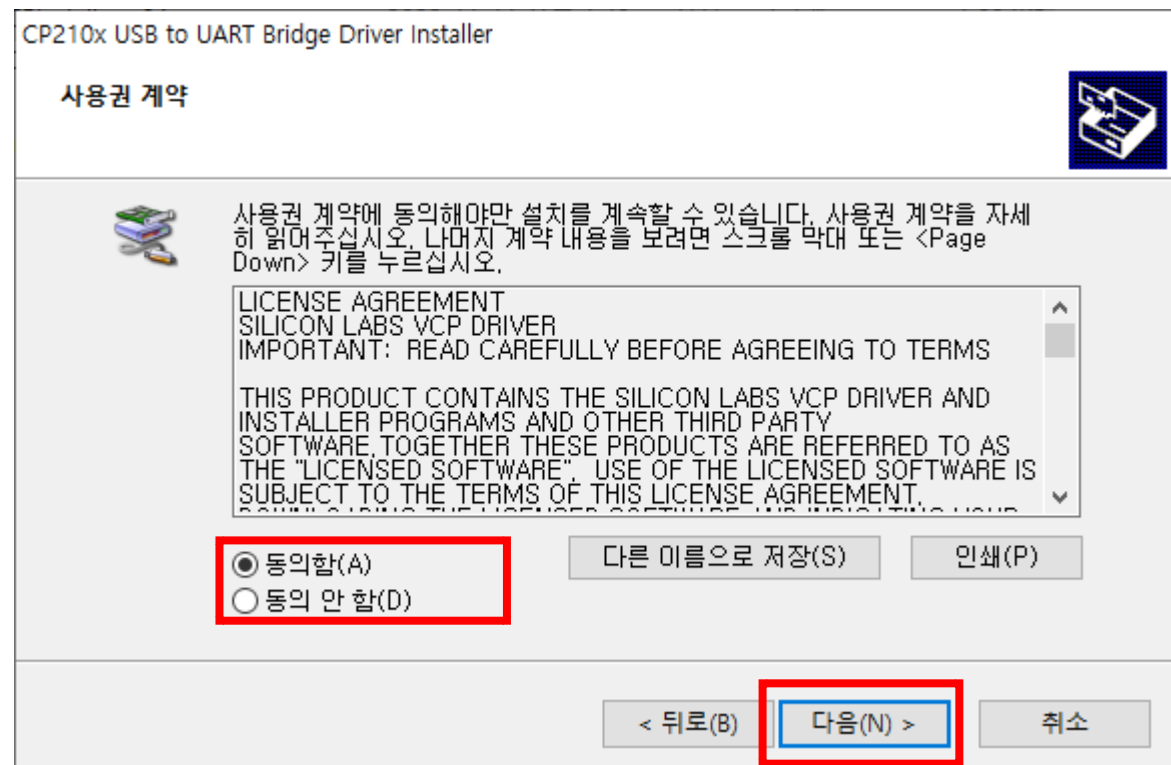
PC 사양에 맞는 프로그램 실행

LiDAR

■ 라이다 드라이버 설치



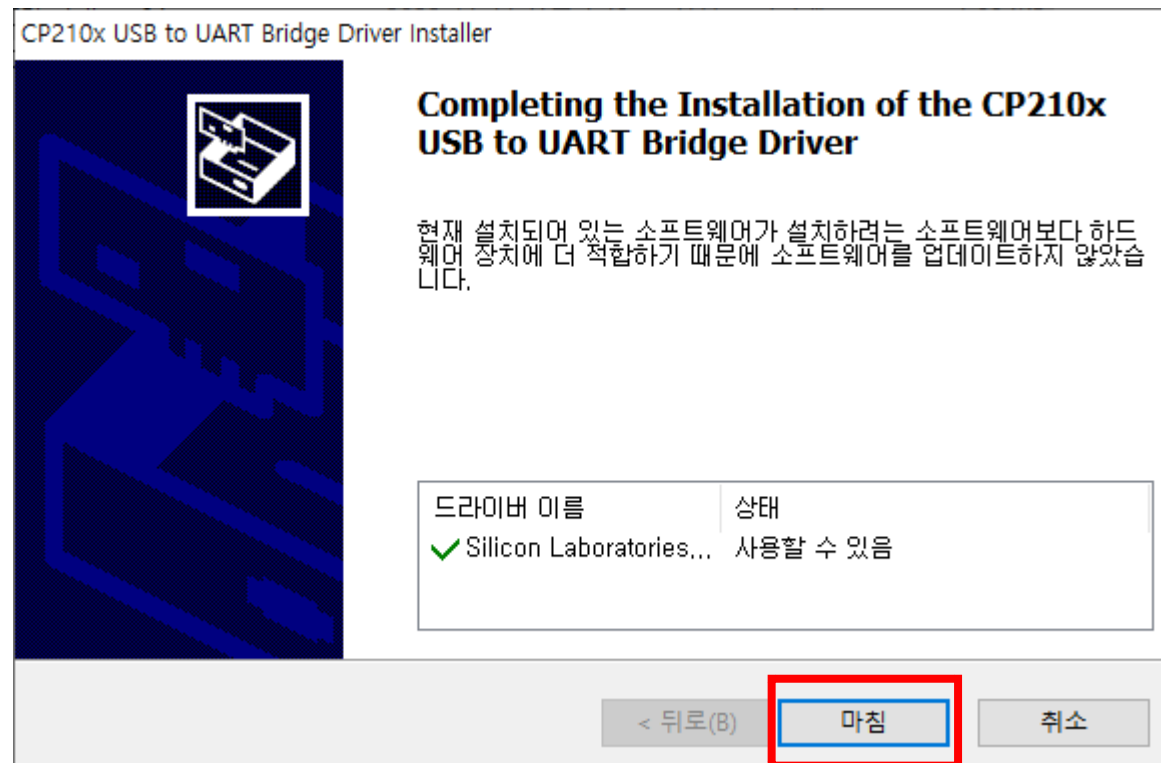
클릭



동의함 체크 후 다음

LiDAR

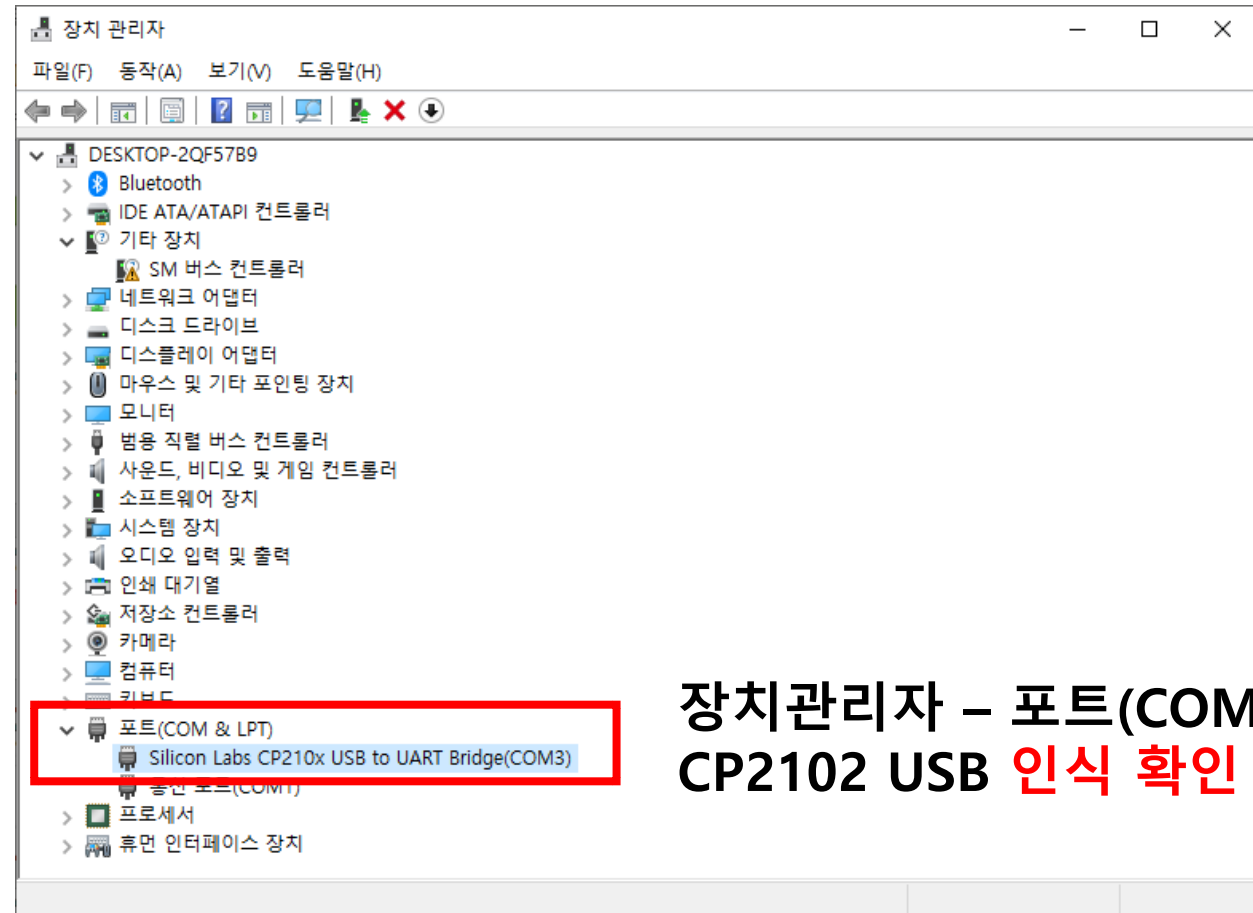
■ 라이다 드라이버 설치



클릭

LiDAR

■ 라이다 장치 인식 확인



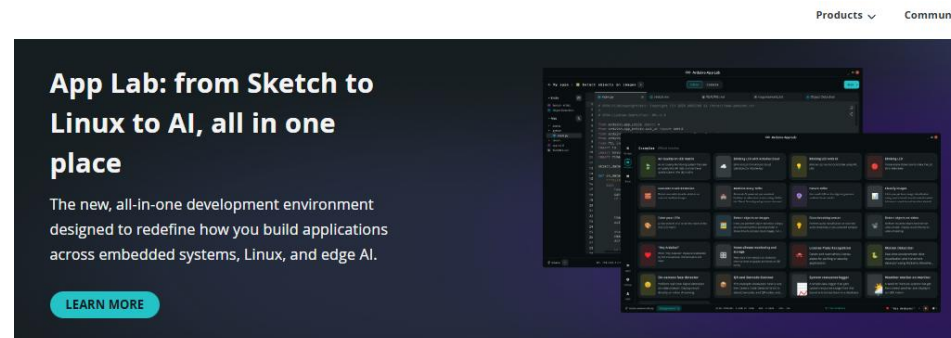
장치관리자 – 포트(COM & LPT)
CP2102 USB 인식 확인

Arduino IDE

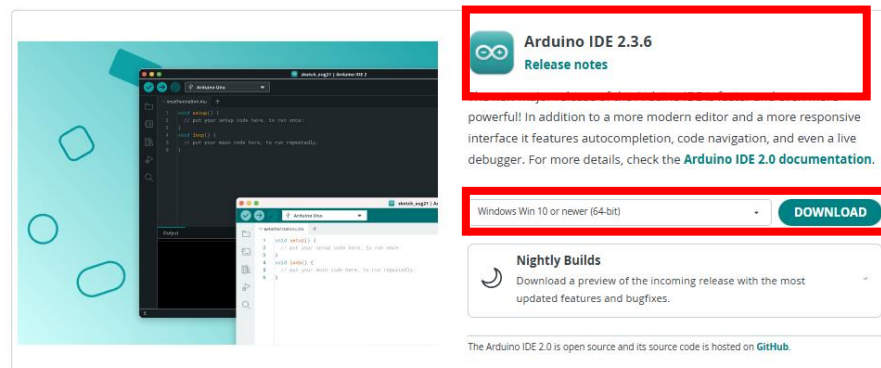
■ Arduino IDE 설치

다운로드 페이지:

<https://www.arduino.cc/en/software/>



Bring Your Projects to Life with Arduino Software

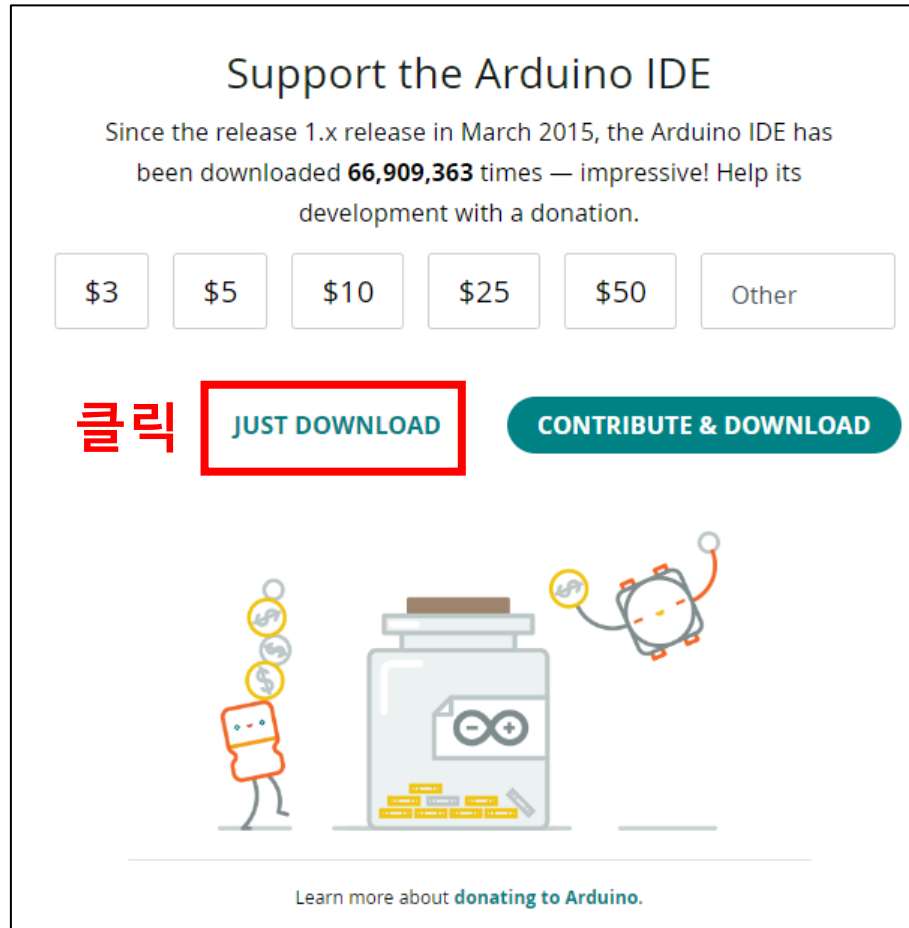


Arduino IDE 2.3.6
버전 확인

Windows Win 10 or newer
클릭 후 다운로드

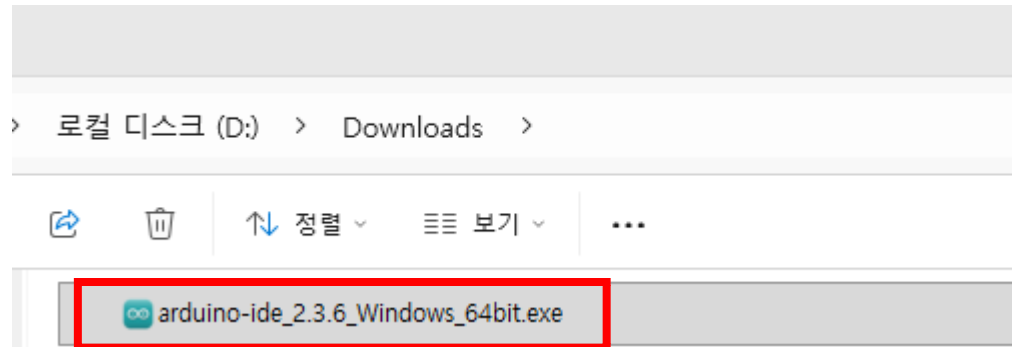
Arduino IDE

■ Arduino IDE 설치

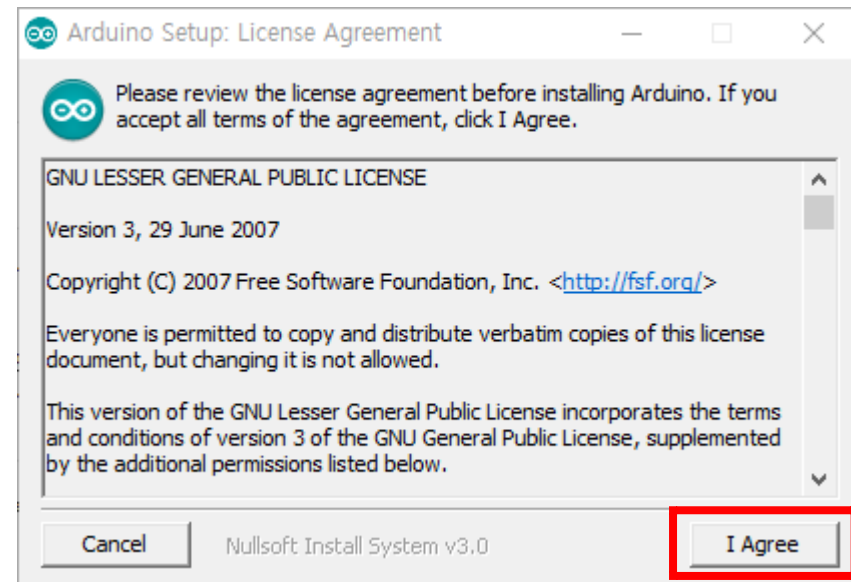


Arduino IDE

■ Arduino IDE 설치



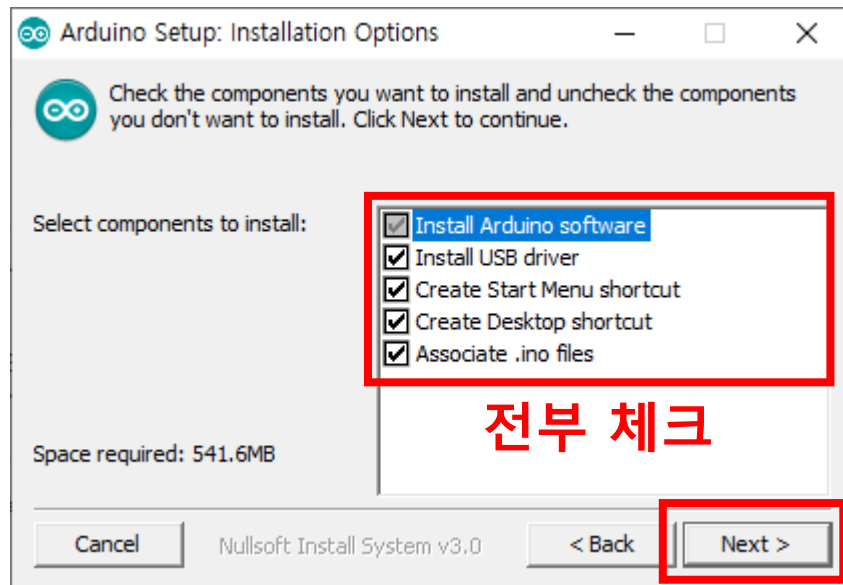
클릭



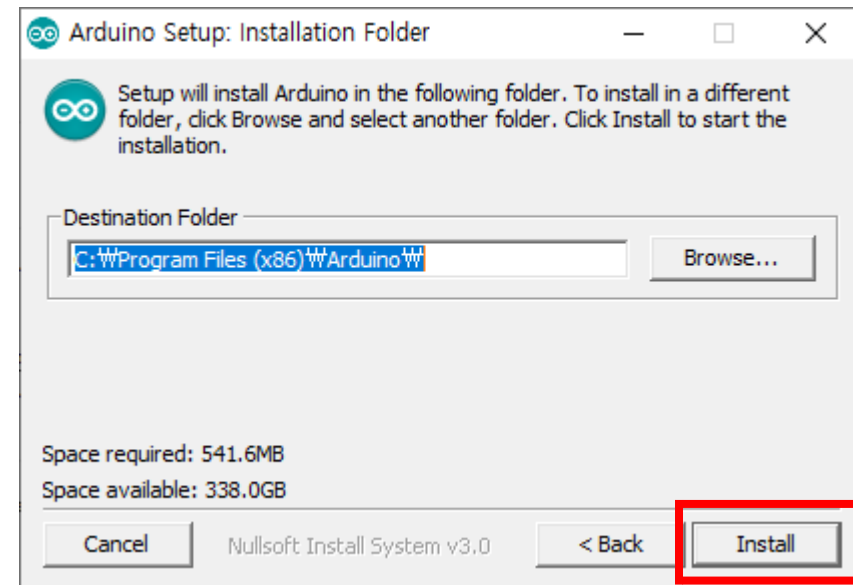
클릭

Arduino IDE

■ Arduino IDE 설치



클릭



클릭

Arduino IDE

■ Arduino IDE 설치

장치 소프트웨어 설치 알림 뜰 시, 모두 설치



Arduino IDE

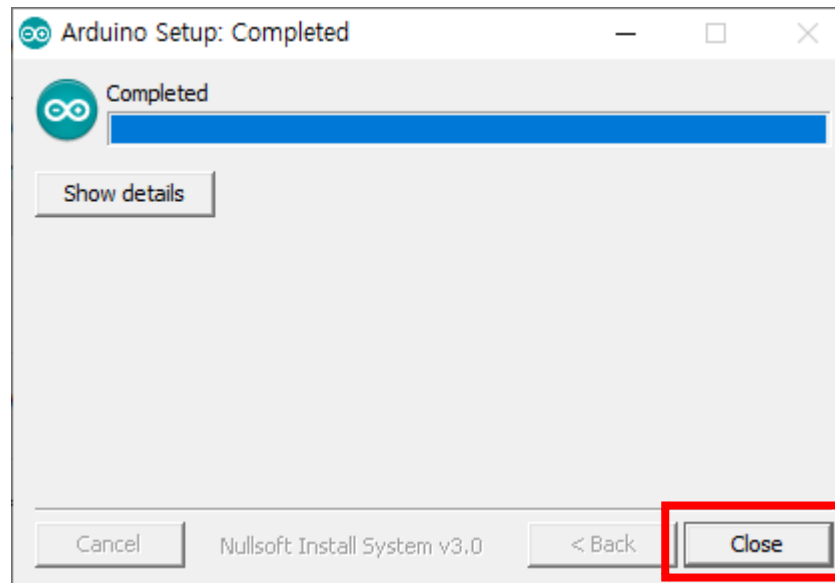
■ Arduino IDE 설치

장치 소프트웨어 설치 알림 뜰 시, 모두 설치



Arduino IDE

■ Arduino IDE 설치

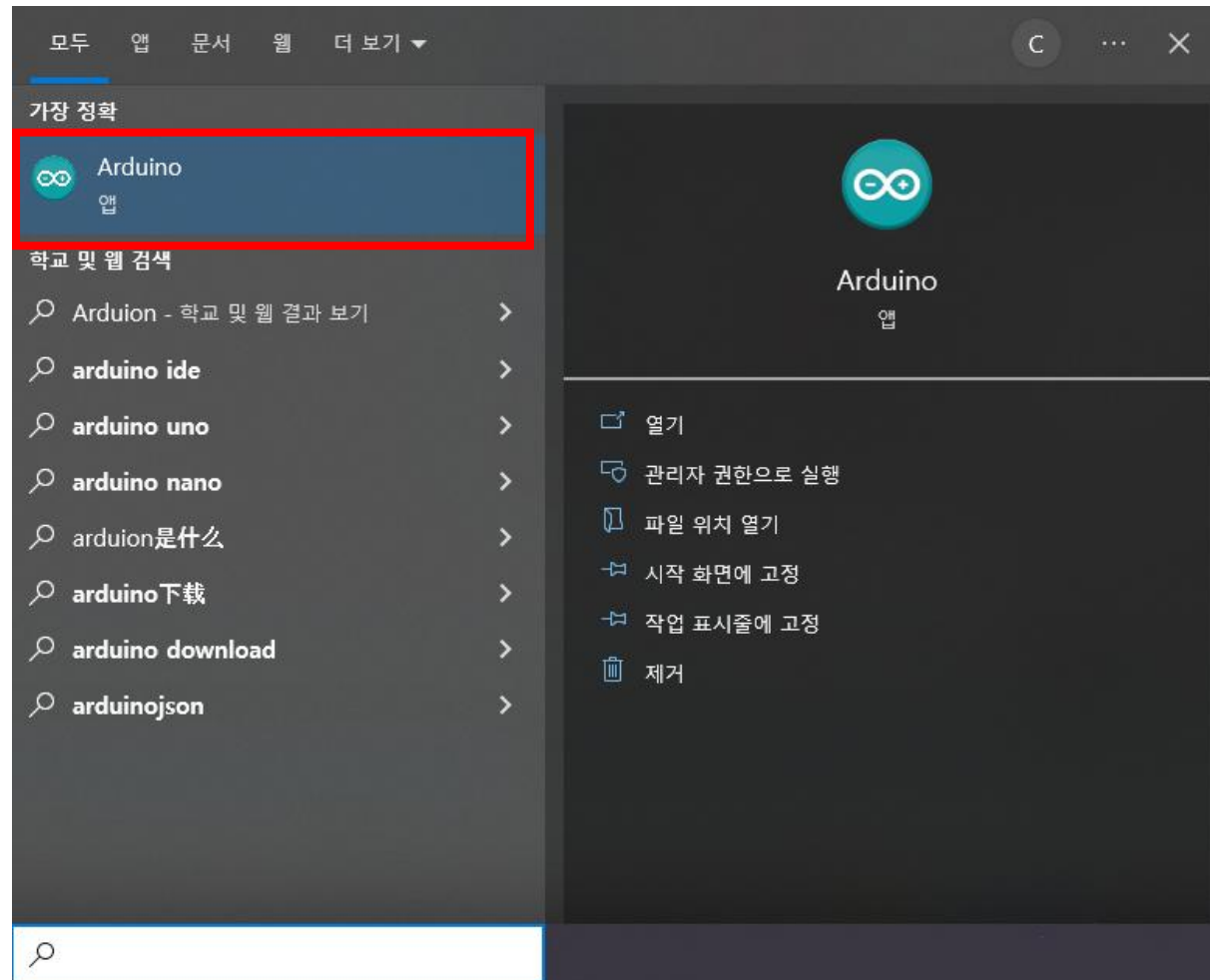


클릭

Arduino IDE

■ Arduino IDE 설치

Arduino 실행



Arduino IDE

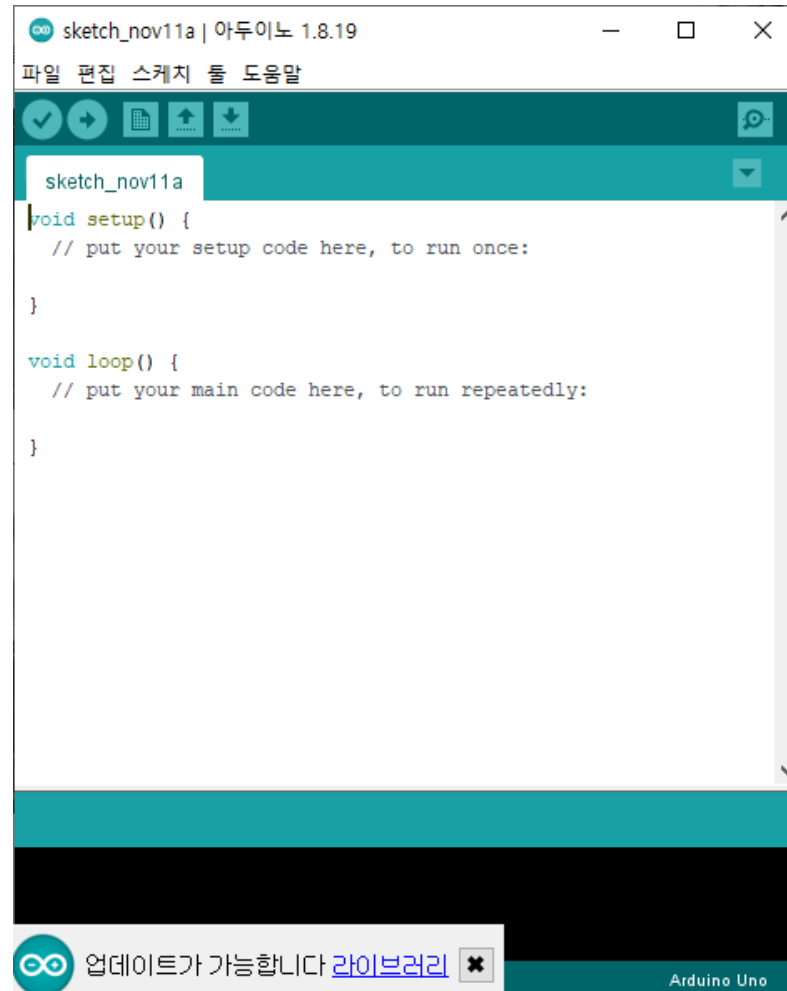
■ Arduino IDE 설치



개인 네트워크 체크

Arduino IDE

■ Arduino IDE 초기 화면



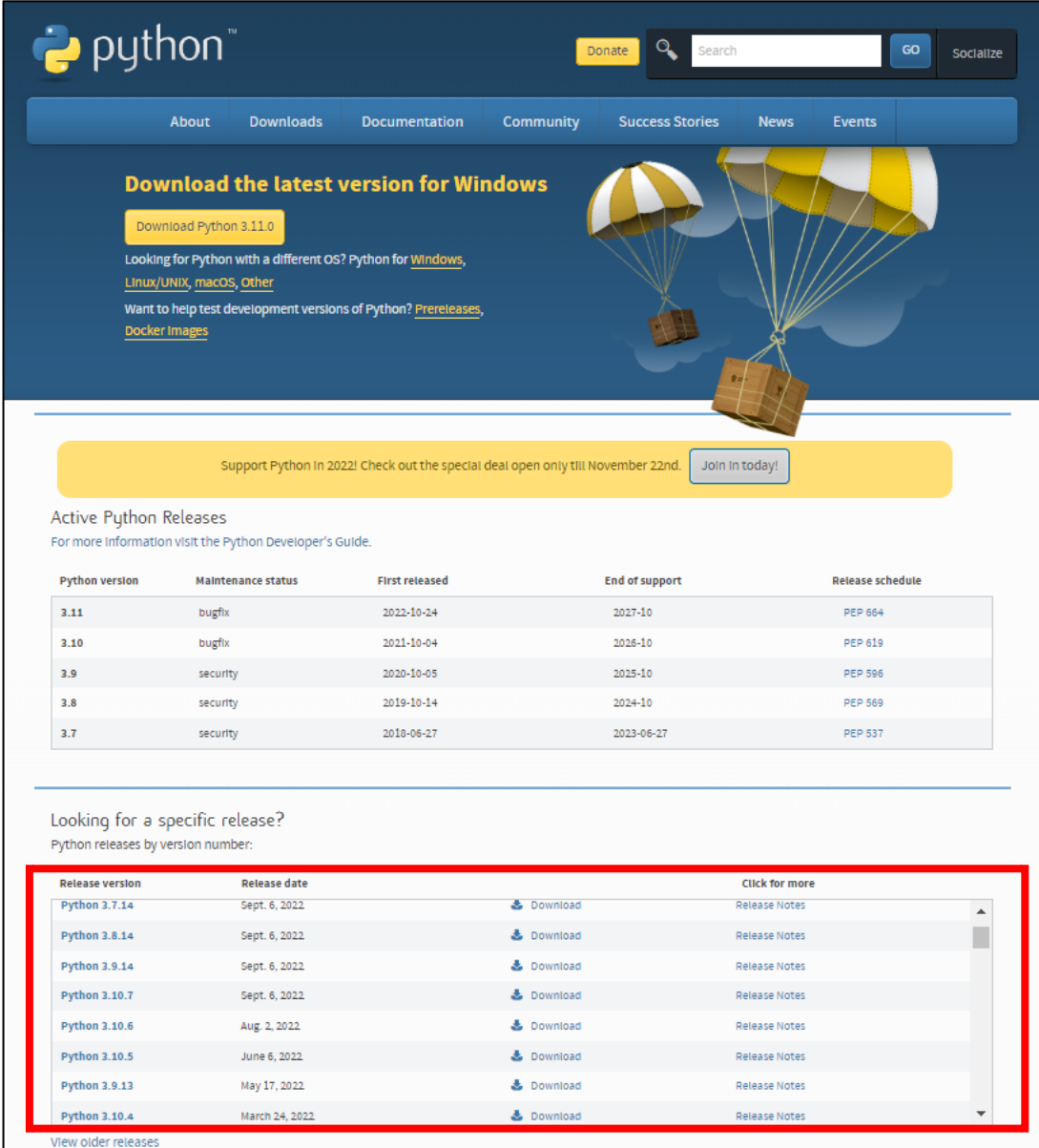
Python

■ Python 설치

다운로드 페이지:

<https://www.python.org/downloads/>

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선택하여 다운로드



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Active Python Releases

For more information visit the Python Developer's Guide.

Python version	Maintenance status	First released	End of support	Release schedule
3.11	bugfix	2022-10-24	2027-10	PEP 664
3.10	bugfix	2021-10-04	2026-10	PEP 619
3.9	security	2020-10-05	2025-10	PEP 596
3.8	security	2019-10-14	2024-10	PEP 569
3.7	security	2018-06-27	2023-06-27	PEP 537

Looking for a specific release?

Python releases by version number:

Release version	Release date	Download	Click for more
Python 3.7.14	Sept. 6, 2022	Download	Release Notes
Python 3.8.14	Sept. 6, 2022	Download	Release Notes
Python 3.9.14	Sept. 6, 2022	Download	Release Notes
Python 3.10.7	Sept. 6, 2022	Download	Release Notes
Python 3.10.6	Aug. 2, 2022	Download	Release Notes
Python 3.10.5	June 6, 2022	Download	Release Notes
Python 3.9.13	May 17, 2022	Download	Release Notes
Python 3.10.4	March 24, 2022	Download	Release Notes

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Python

■ Python 설치

다운로드 페이지:

<https://www.python.org/downloads/>

Release version	Release date	Click for more	
Python 3.7.14	Sept. 6, 2022	Download	Release Notes
Python 3.8.14	Sept. 6, 2022	Download	Release Notes
Python 3.9.14	Sept. 6, 2022	Download	Release Notes
Python 3.10.7	Sept. 6, 2022	Download	Release Notes
Python 3.10.6	Aug. 2, 2022	Download	Release Notes
Python 3.10.5	June 6, 2022	Download	Release Notes
Python 3.9.13	May 17, 2022	Download	Release Notes
Python 3.10.4	March 24, 2022	Download	Release Notes

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본 교육에서는 **3.9.13 버전**을 활용합니다.
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Python 3.9.13

Release Date: May 17, 2022

This is the thirteenth and final regular maintenance release of Python 3.9

Note: The release you're looking at is Python 3.9.13, a **bugfix release** for the legacy 3.9 series. Python 3.10 is now the latest feature release series of Python 3. [Get the latest release of 3.10 a free](#).

According to the release calendar specified in [PEP 556](#), Python 3.9.13 is the final regular maintenance release. Starting now, the 3.9 branch will only accept security fixes, and releases of those will be made in source-only form until October 2026.

Compared to the 3.8 series, this last regular bugfix release is still pretty active at 266 commits since 3.9.12. In comparison, version 3.8.20, the final regular bugfix release of Python 3.8, included only 10 commits, however, it's likely that it was 3.8 that was special here with the governance changes occupying core developers' minds. For reference, version 3.7.16, the final regular bugfix release of Python 3.7, included 187 commits.

In any case, 266 commits is quite a few changes, some of which being pretty important fixes. Take a look at the [change log](#) for details.

Major new features of the 3.9 series, compared to 3.8

Some of the new major new features and changes in Python 3.9 are:

- [PEP 569](#), Module State Access from C: Extension Methods
- [PEP 568](#), Union Operators in C (C4C)
- [PEP 565](#), Type Hinting Services in Standard Collections
- [PEP 563](#), Iterable Function and variable annotations
- [PEP 560](#), Python adopts a stable annual release cadence
- [PEP 555](#), Relaxing Grammar Restrictions On decorators
- [PEP 553](#), Support for the IANA Time zone Database in the Standard Library
- [PEP 550](#), String methods to remove prefixes and suffixes
- [PEP 547](#), New PEP parser for CPython
- [PEP 546](#), garbage collection does not block on unconnected objects
- [PEP 545](#), `os.path` again added that allows process management without races and signals
- [PEP 540](#), Unicode support updated to version 10.0.0
- [PEP 535](#), when Python is initialized multiple times in the same process, it does not leak memory anymore
- A number of Python built-ins (`range`, `tuple`, `set`, `dict`) are now speed up using [PyPy](#) bytecode
- A number of Python modules (`_abc`, `audioop`, `_codecs`, `_codecs_cn`, `_codecs_iso2022`, `_crypt`, `_functools`, `_hash`, `_locale`, `operator`, `re`, `random`, `time`, `weakref`) now use [multiphase](#) installation as defined by [PEP 544](#)
- A number of standard library modules (`audioop`, `set`, `sys`, `_hashlib`, `pwd`, `_posixsubprocess`, `random`, `select`, `struct`, `termios`, `zlib`) are now using the stable ABI defined by [PEP 544](#)

You can find a more comprehensive list in this release's ["What's New"](#) document.

More resources

- [Online Documentation](#)
- [PEP 560](#), 3.9 Release Schedule
- Report bugs at <https://bugs.python.org>
- [Help: Start Python and its community](#)

And Now for Something Completely Different

MC Python (John Cleese): "Well, PEEPER! (holding) (singing) This is your new o'clock alarm call! (takes point out of the cage), throws it up in the air and watches it plummet to the floor.)

MC Python: Now that's what I call a dead parrot.

Durwin (Michael Palin): No, no...No, he's retired!

MC Python: I UNDERST!

Durwin: Well, 'ow stumped him, just as he was walkin' up Norwagian Alms that rainy, rainy.

MC Python: Ah... now look, man... I'm definitely 'ot enough of this. That parrot is definitely deceased, and when I purchased it, not 'alf an hour ago, you assured me that its total lack of movement was due to it being 'tired and shaggy and following a prolonged seizure.

Durwin: Well, he's... he's... ah... probably pining for the Spain.

[Full Changelog](#)

Files

Version	Operating System	Description	MD5 Sum	File Size	GPU
Original source (tarball)		Source release	af76c081f8b12171c3a0f0c0e0f2ab17	291516087	Size
32 compressed source (tarball)		Source release	6a011212170b00002030f020b122020b	291516088	Size
macOS bin for Intel-only install	macOS	for macOS 10.16 and later, deprecated	67148000000000000000000000000000	309611241	Size
macOS bin for Universal2 install	macOS	for macOS 10.16 and later	70b0a702000000000000000000000000	308021242	Size
Windows installer (32-bit)	Windows		500b0000000000000000000000000000	7729900	Size
Windows installer (64-bit)	Windows		37713170001100000000000000000000	86707100	Size
Windows Help file	Windows		00000000000000000000000000000000	8900000	Size
Windows installer (32-bit)	Windows		00000000000000000000000000000000	200000000	Size
Windows installer (64-bit)	Windows	Not recommended	00000000000000000000000000000000	292000000	Size

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Python Conference

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Python Newsletter

Community News

Python News

Python News

Scientific

창 아래쪽 Files 확인

 Automation Lab.

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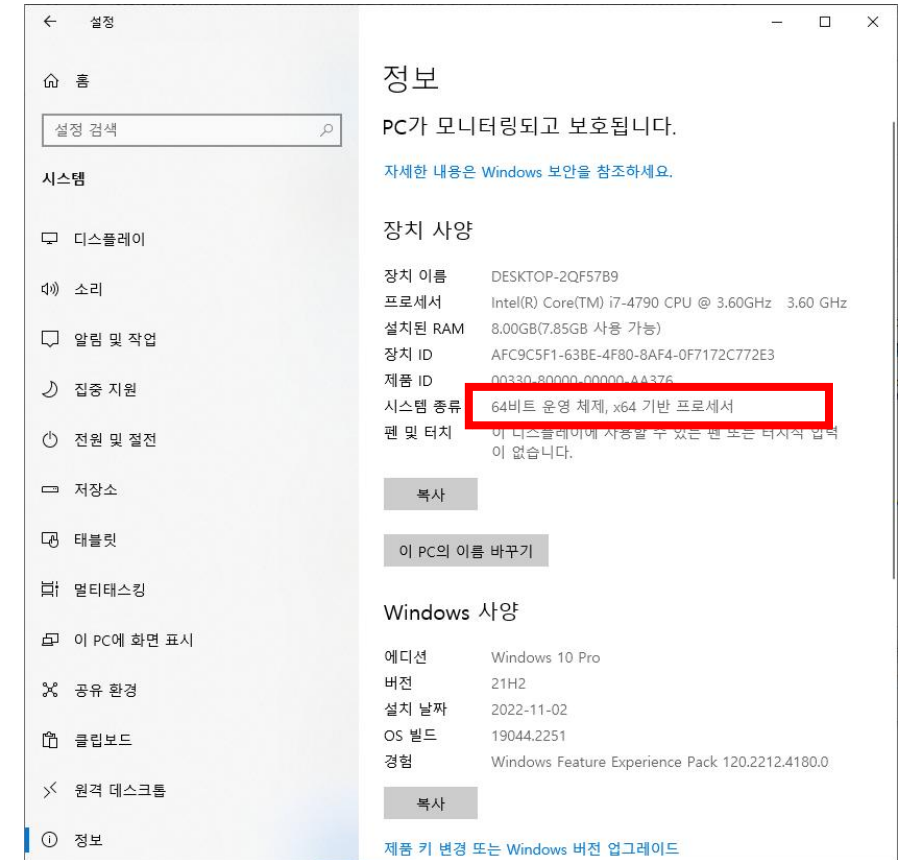
Python

■ Python 설치

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		eafda83543bad127cadef4d288fdab87	26355887	SIG
XZ compressed source tarball	Source release		5e2411217b0060828d5f923eb422a3b8	19754368	SIG
macOS 64-bit Intel-only installer	macOS	for macOS 10.9 and later, deprecated	671848930809decf27f586ddf98c6e9b	30997161	SIG
macOS 64-bit universal2 installer	macOS	for macOS 10.9 and later	76b63cf623e32cdf27c5033434bd69ce	38821163	SIG
Windows embeddable package (32-bit)	Windows		fec0bc06857502a56dd1aeaea6488ef8	7729405	SIG
Windows embeddable package (64-bit)	Windows		57731cf80b1c429a0be7133266d7d7cf	8570740	SIG
Windows help file	Windows		c86feba059b340a1de2a9d2ee7059a6d	8953644	SIG
Windows installer (32-bit)	Windows		46c35b0a2a4325c275b2ed3187b08ac4	28096840	SIG
Windows installer (64-bit)	Windows	Recommended	e7062b85c3624af82079794729618eca	29235432	SIG

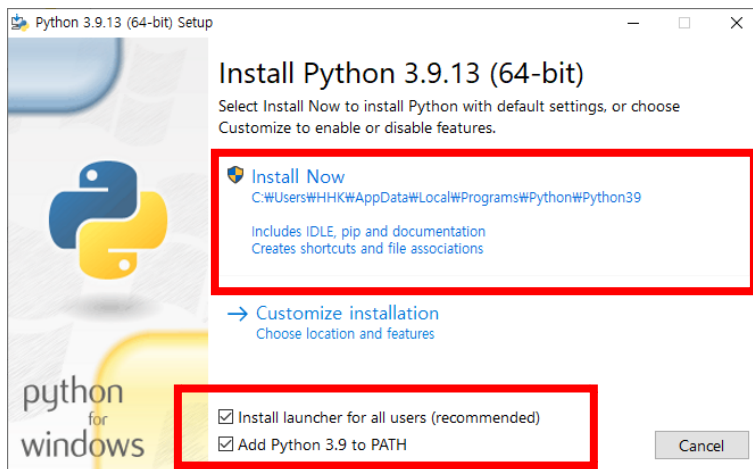
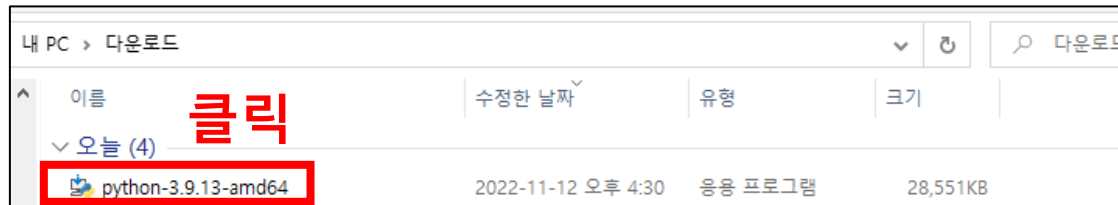
본인 PC 하드웨어 사양(64bit, 32bit) 선택



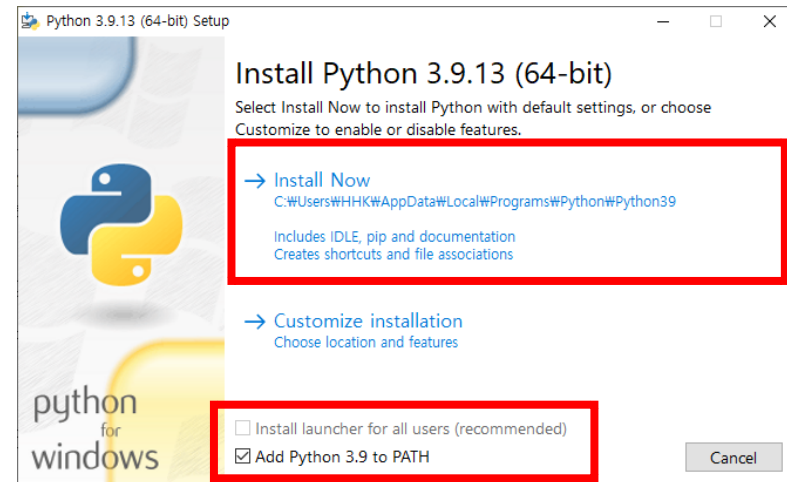
하드웨어 사양
본 교육자료 첫 부분(page 5)에서 확인

Python

■ Python 설치



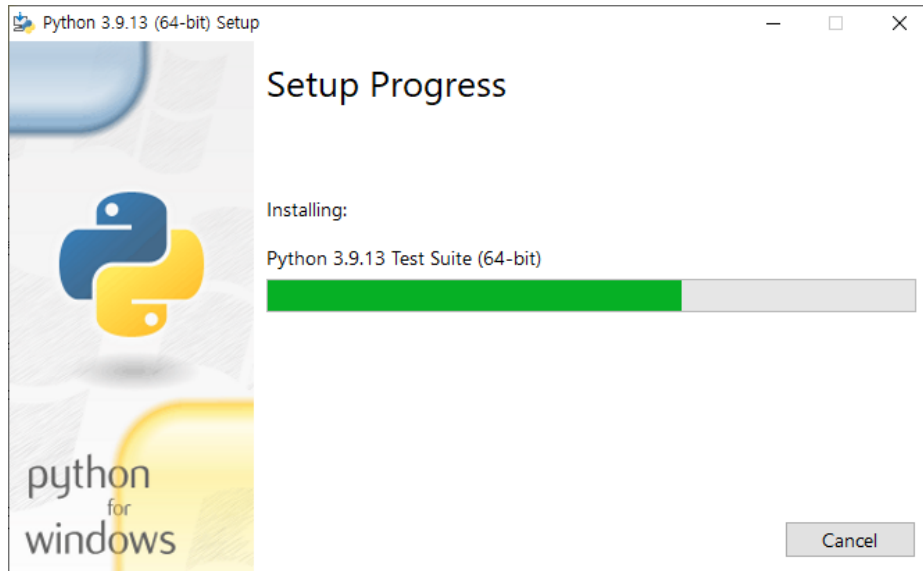
체크박스 둘 다 활성화 되어 있는 경우
둘 다 체크



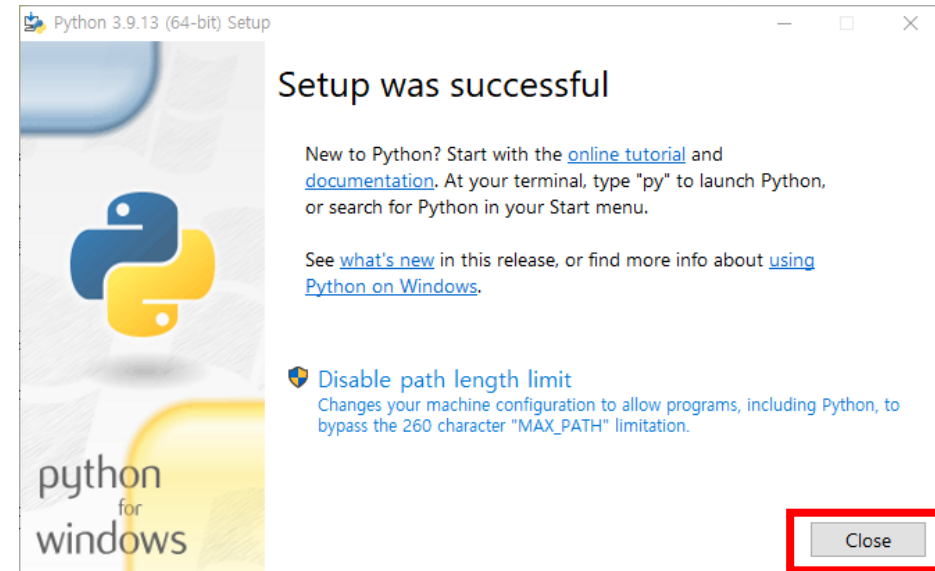
하나만 활성화 되어있는 경우
아래 것만 체크

Python

■ Python 설치



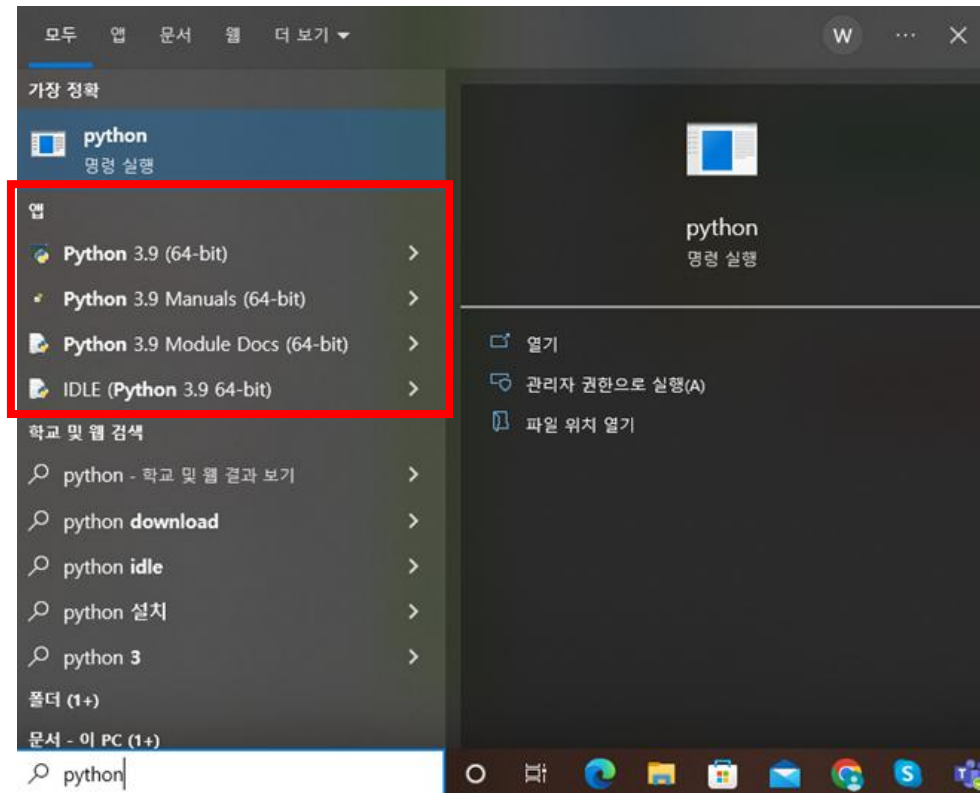
설치 중 화면



클릭

Python

■ Python 설치



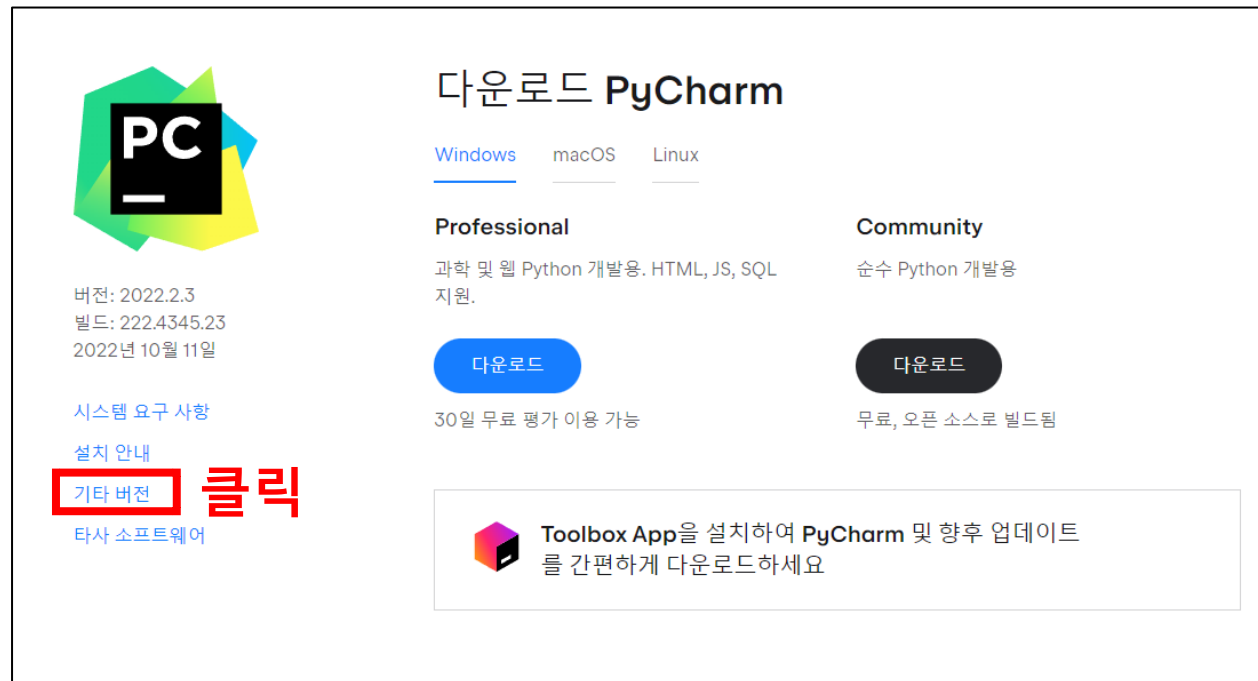
Python 검색 후
설치 완료되었는지 확인

PyCharm

■ PyCharm 설치

다운로드 페이지:

<https://www.jetbrains.com/ko-kr/pycharm/download/#section=windows>



다운로드 PyCharm

Windows macOS Linux

Professional
과학 및 웹 Python 개발용. HTML, JS, SQL 지원.
다운로드
30일 무료 평가 이용 가능

Community
순수 Python 개발용
다운로드
무료, 오픈 소스로 빌드됨

버전: 2022.2.3
빌드: 222.4345.23
2022년 10월 11일

시스템 요구 사항
설치 안내
기타 버전 클릭
타사 소프트웨어

Toolbox App을 설치하여 PyCharm 및 향후 업데이트를 간편하게 다운로드하세요

PyCharm

■ PyCharm 설치

기타 버전

Version 2022.2

PyCharm Professional Edition

[2022.2.3 - Linux \(tar.gz\)](#)

[2022.2.3 - Windows \(exe\)](#)

[2022.2.3 - macOS \(dmg\)](#)

[2022.2.3 - macOS Apple Silicon \(dmg\)](#)

PyCharm Community Edition

[2022.2.3 - Linux \(tar.gz\)](#)

[2022.2.3 - Windows \(exe\)](#)

[2022.2.3 - macOS \(dmg\)](#)

[2022.2.3 - macOS Apple Silicon \(dmg\)](#)

Version 2022.1

PyCharm Professional Edition

[2022.1.4 - Linux \(tar.gz\)](#)

[2022.1.4 - Windows \(exe\)](#)

[2022.1.4 - macOS \(dmg\)](#)

[2022.1.4 - macOS Apple Silicon \(dmg\)](#)

PyCharm Community Edition

[2022.1.4 - Linux \(tar.gz\)](#)

[2022.1.4 - Windows \(exe\)](#)

[2022.1.4 - macOS \(dmg\)](#)

[2022.1.4 - macOS Apple Silicon \(dmg\)](#)

클릭

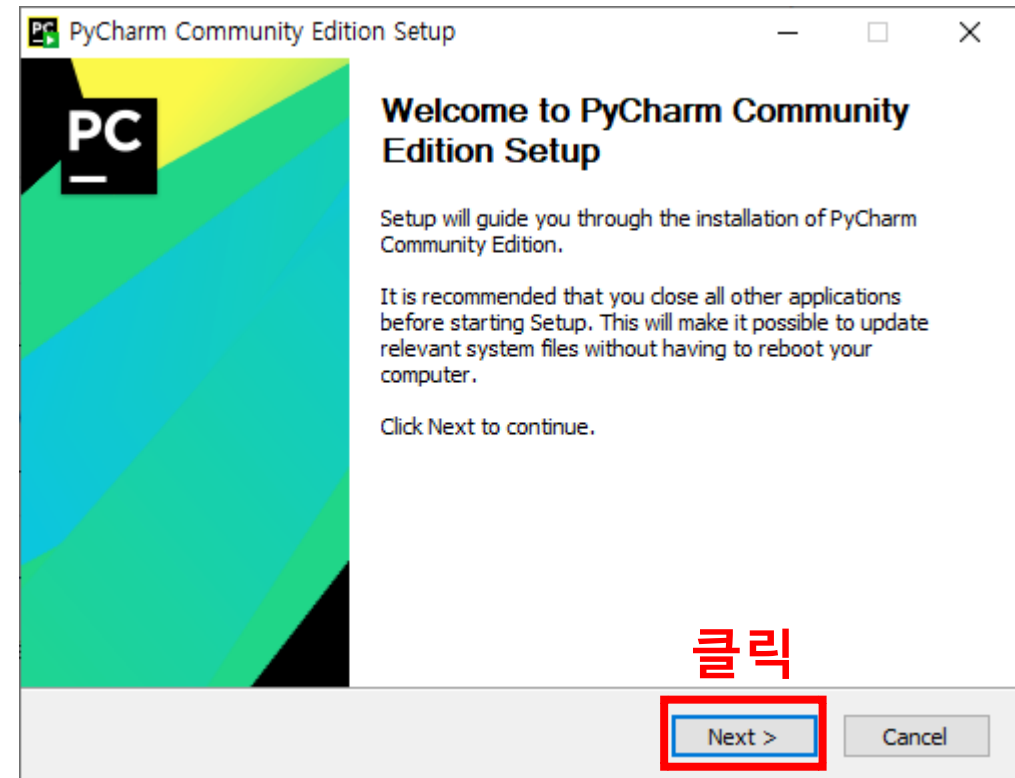
PyCharm

■ PyCharm 설치

PC > 다운로드

이름	수정된 날짜	유형	크기
pycharm-community-2022.1.4	2022-11-12 오후 6:03	응용 프로그램	387,723KB

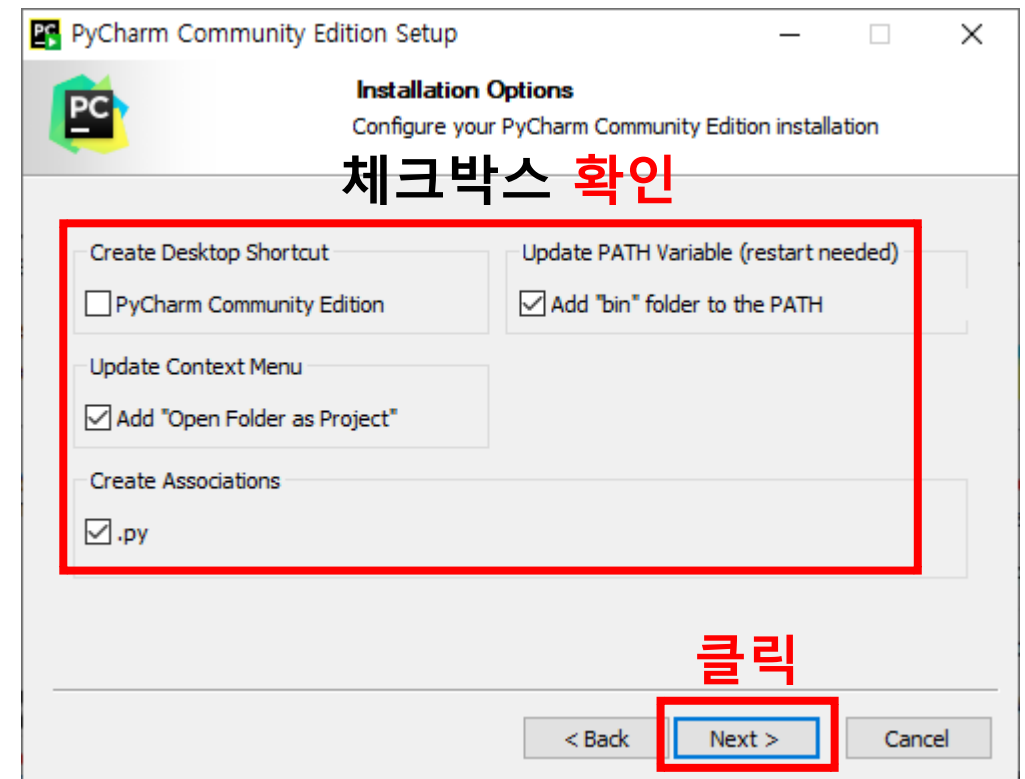
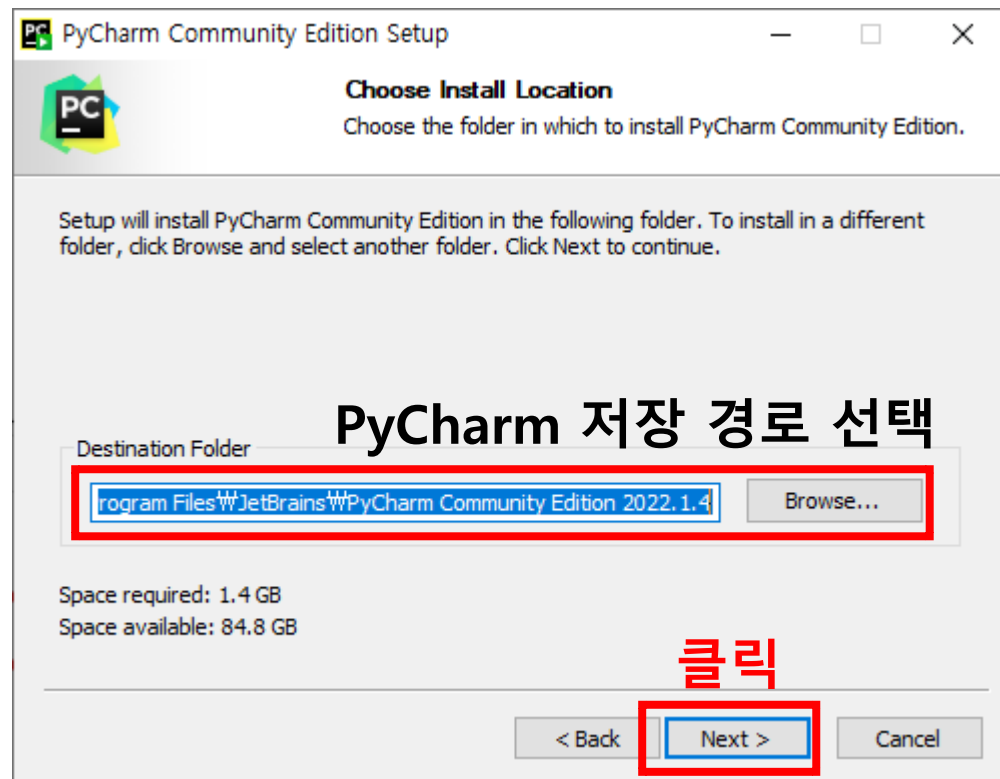
클릭



클릭

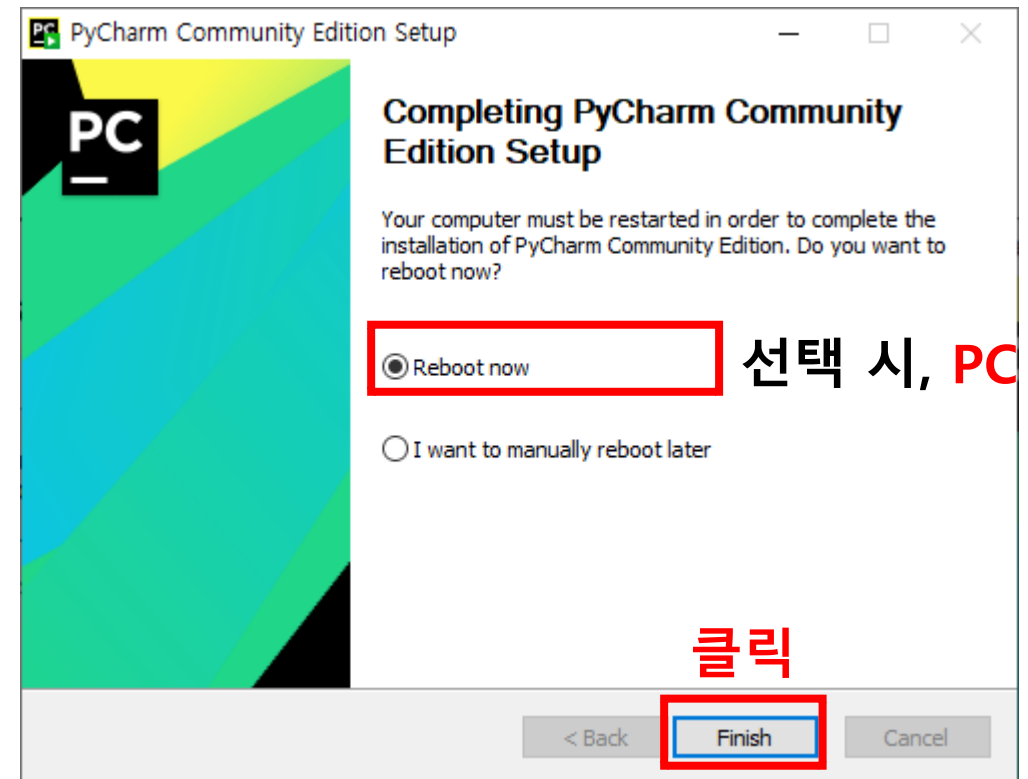
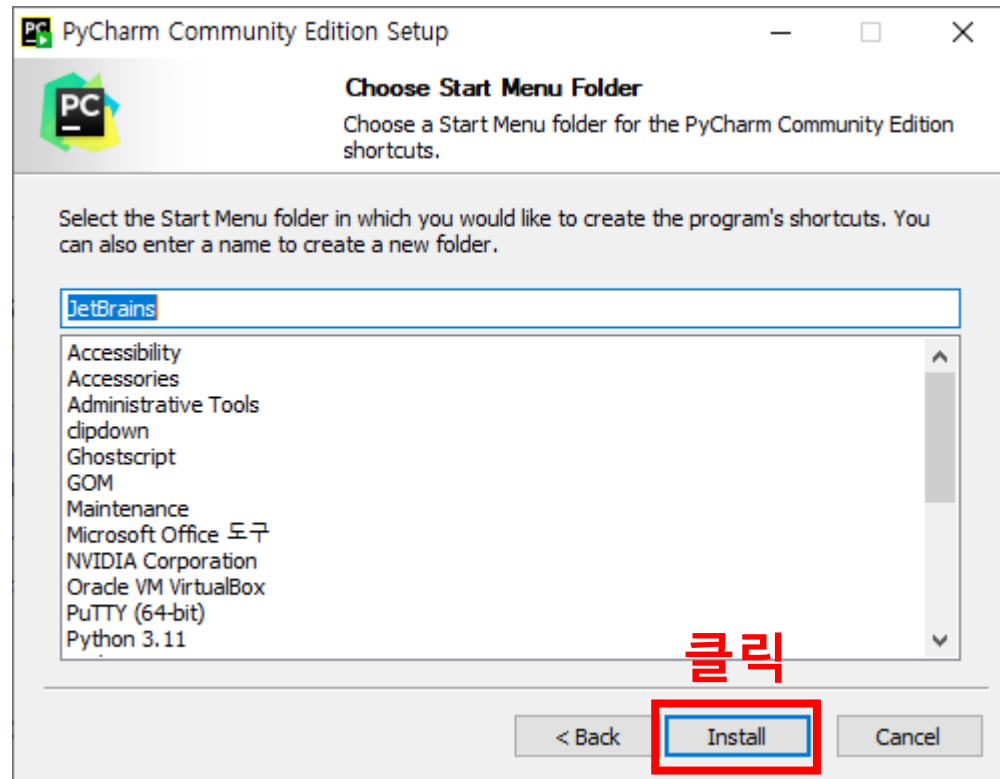
PyCharm

■ PyCharm 설치



PyCharm

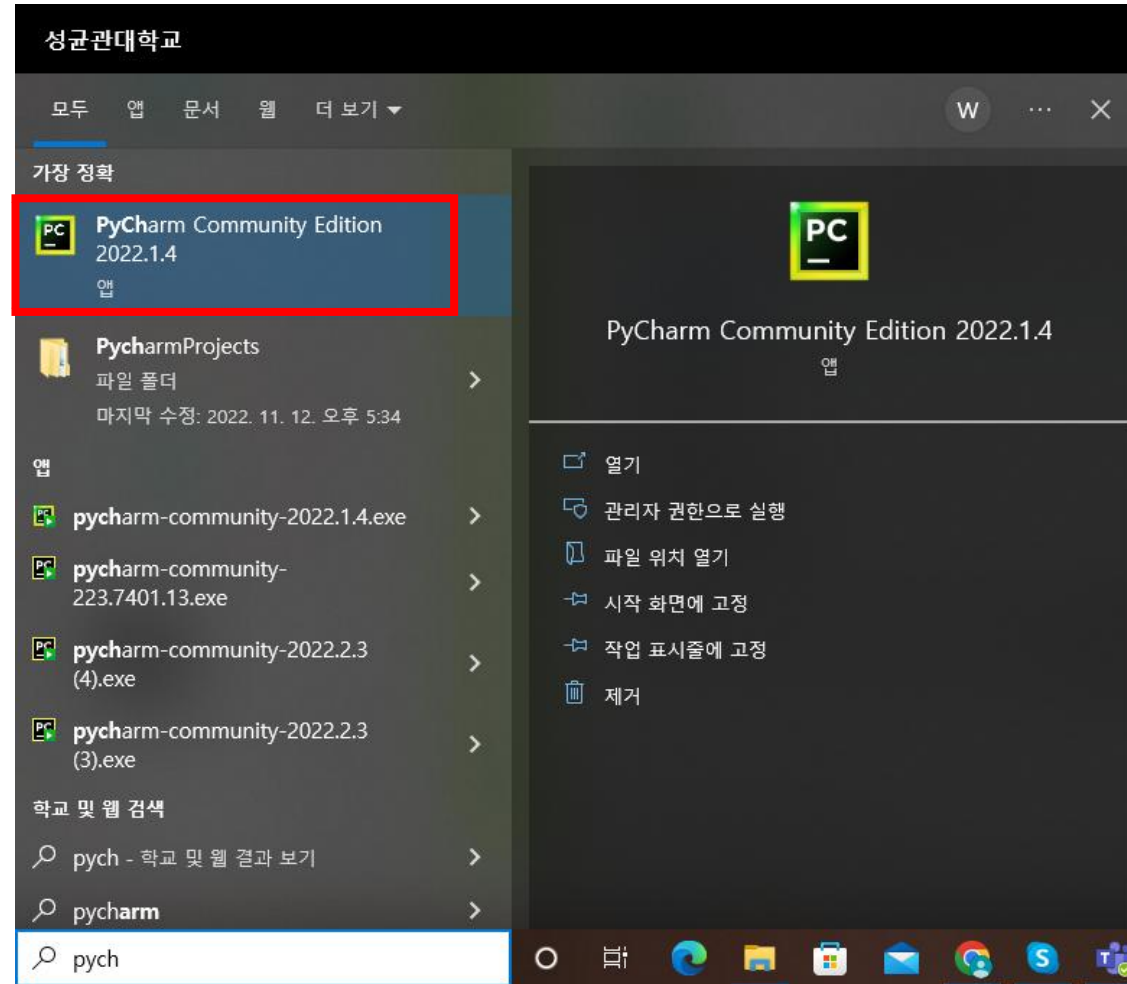
■ PyCharm 설치



PyCharm

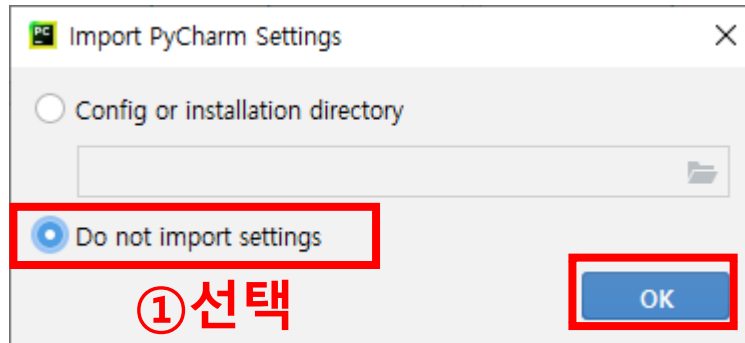
■ Interpreter 설정

PyCharm 실행



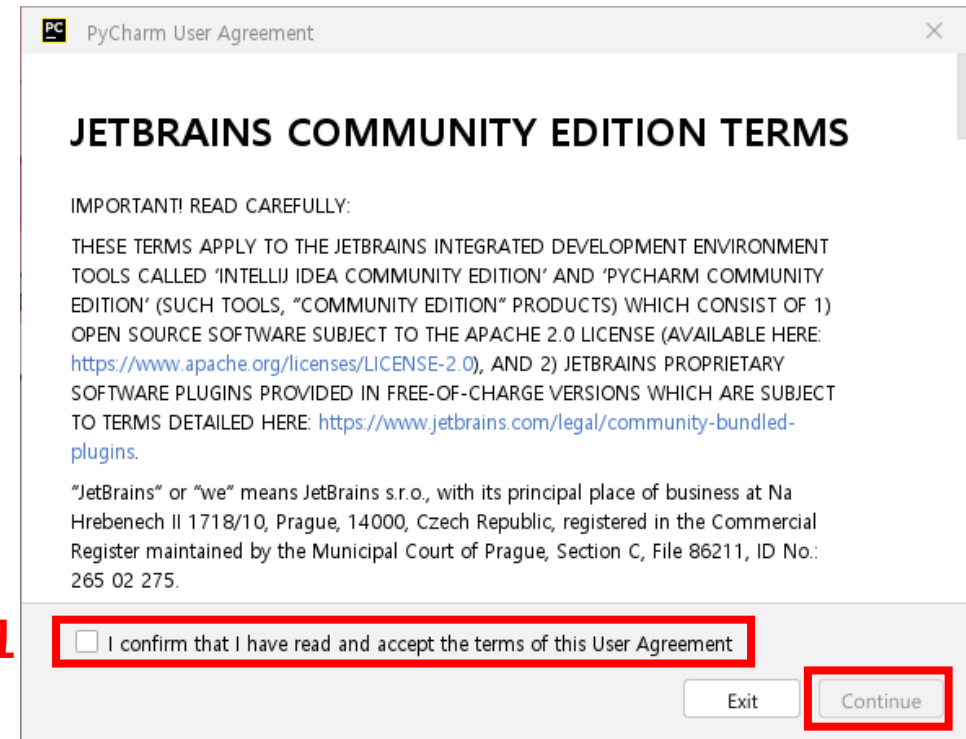
PyCharm

■ Interpreter 설정



① 선택

② 클릭

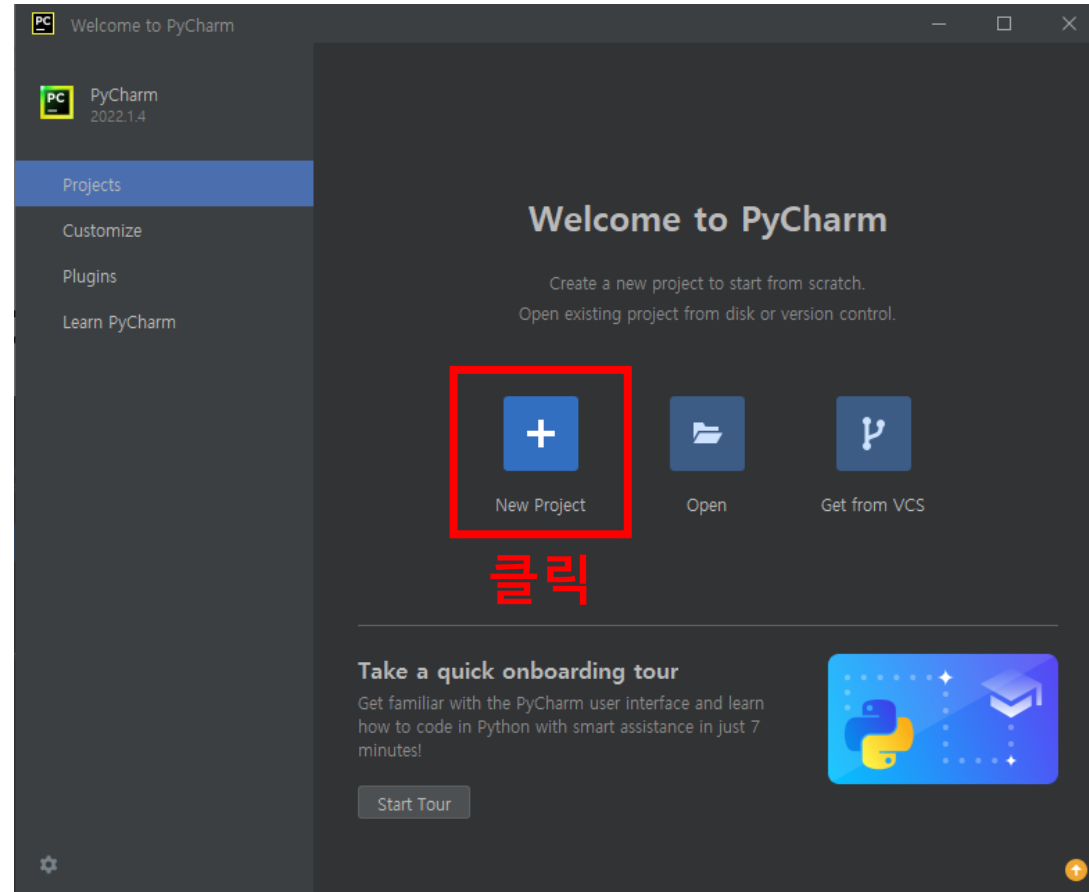


① 체크

② 클릭

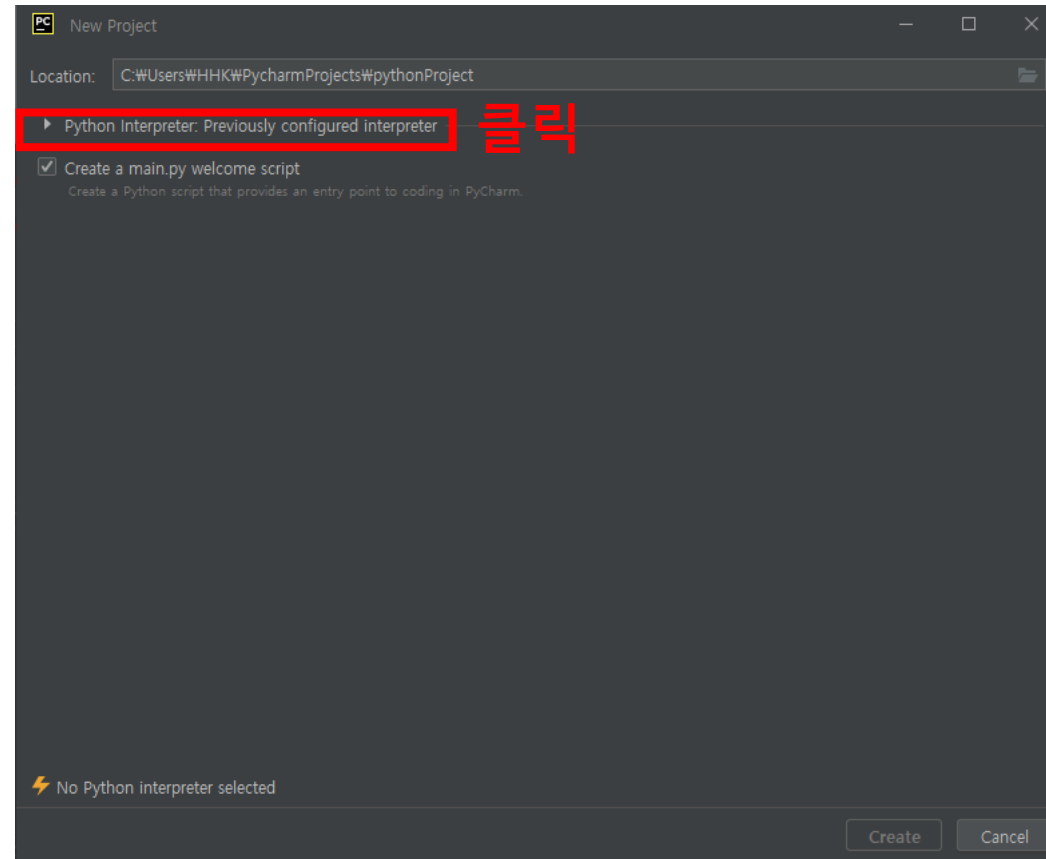
PyCharm

■ Interpreter 설정



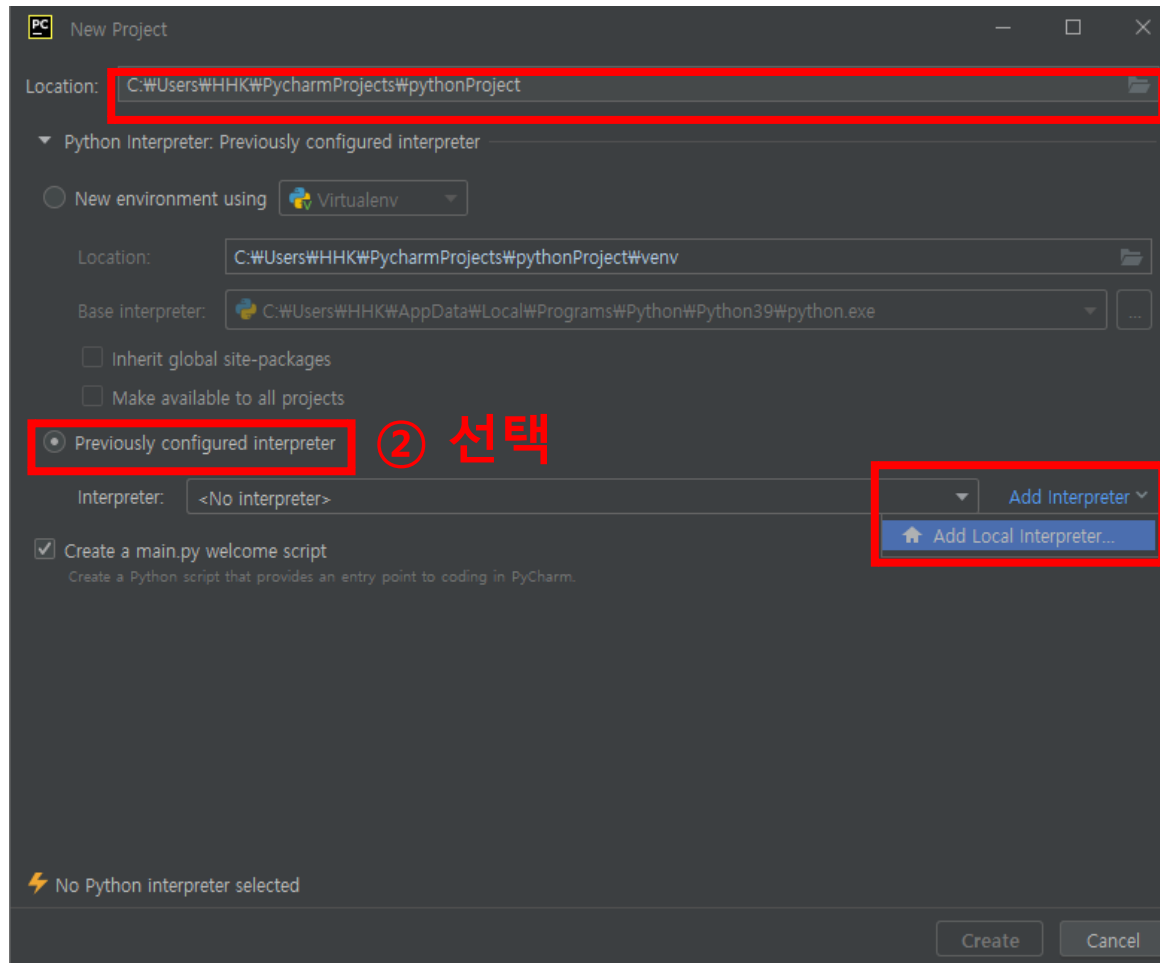
PyCharm

■ Interpreter 설정



PyCharm

■ Interpreter 설정



① 프로젝트 파일 폴더 위치 지정
(초기값 유지)

③ Add Interpreter
-> Add Local Interpreter 클릭

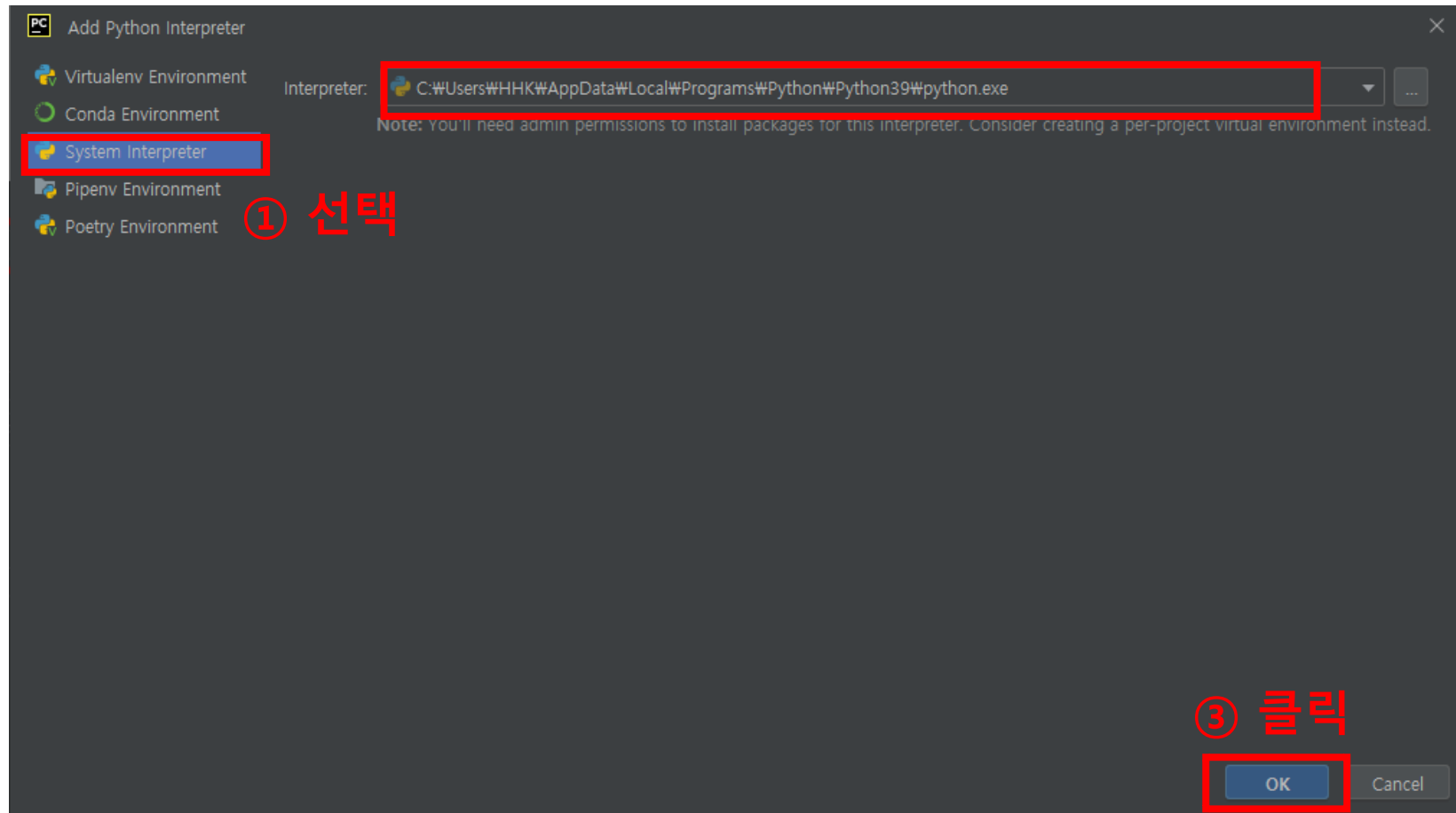


③-1 다음과 같이 뜨는 경우 ... 클릭하여 진행

PyCharm

■ Interpreter 설정

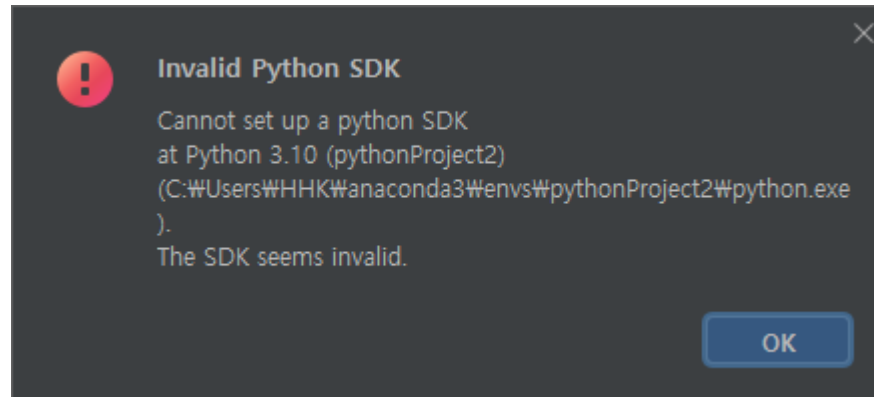
② Interpreter 선택



만일 Interpreter 없으면,
Python이 설치 안 된 것
Python 재설치 필요

PyCharm

■ Interpreter 설정

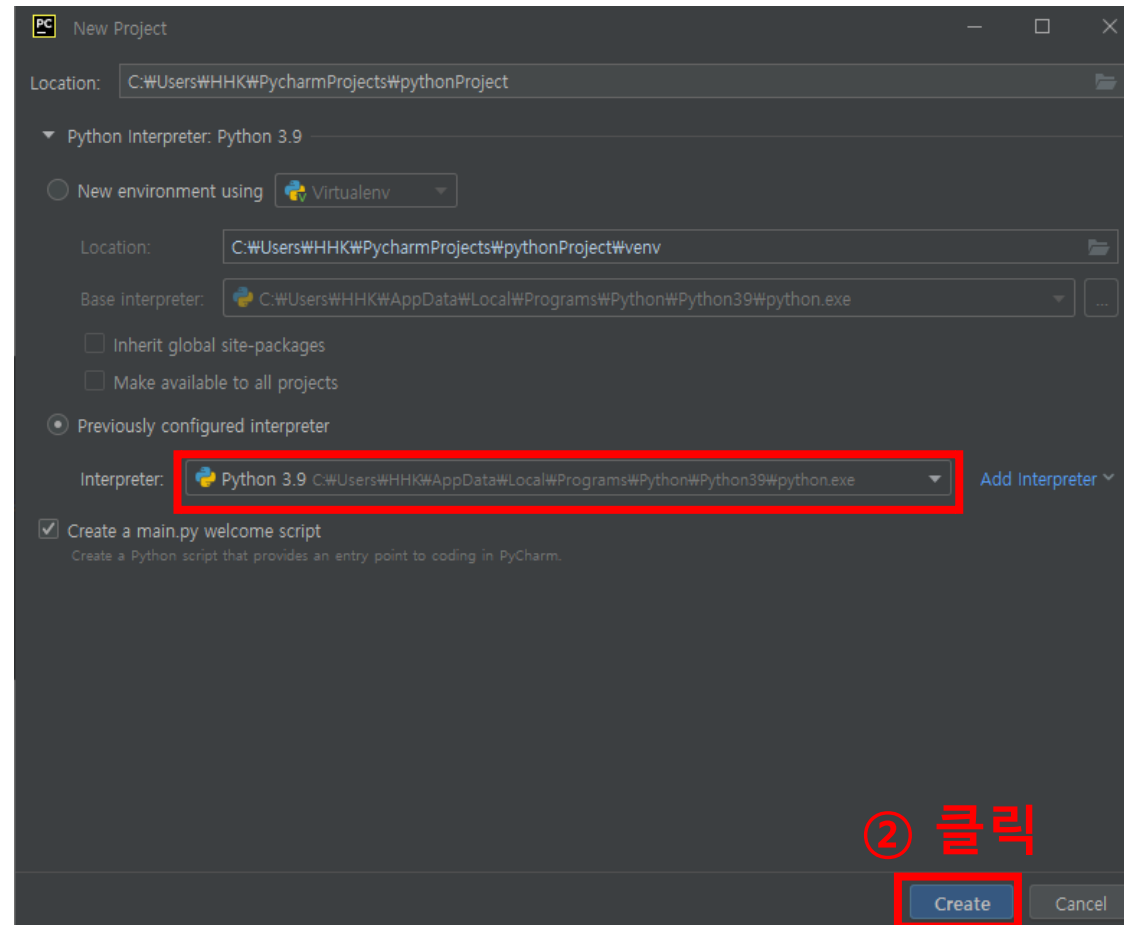


다음과 같은 오류 발생 시

2022.1.4 버전으로 **재설치**

PyCharm

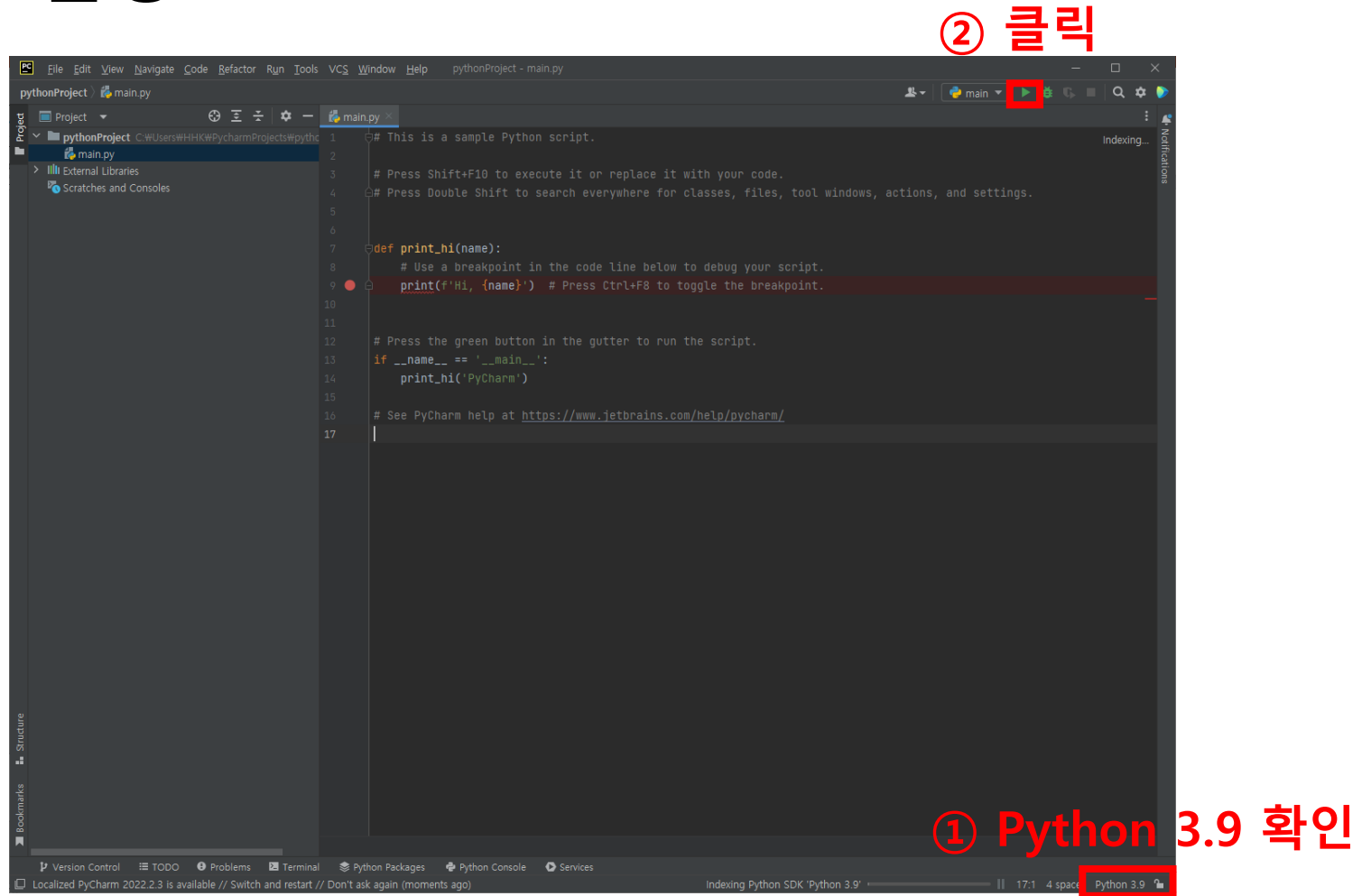
■ Interpreter 설정



① interpreter 확인
(Python 3.9)

PyCharm

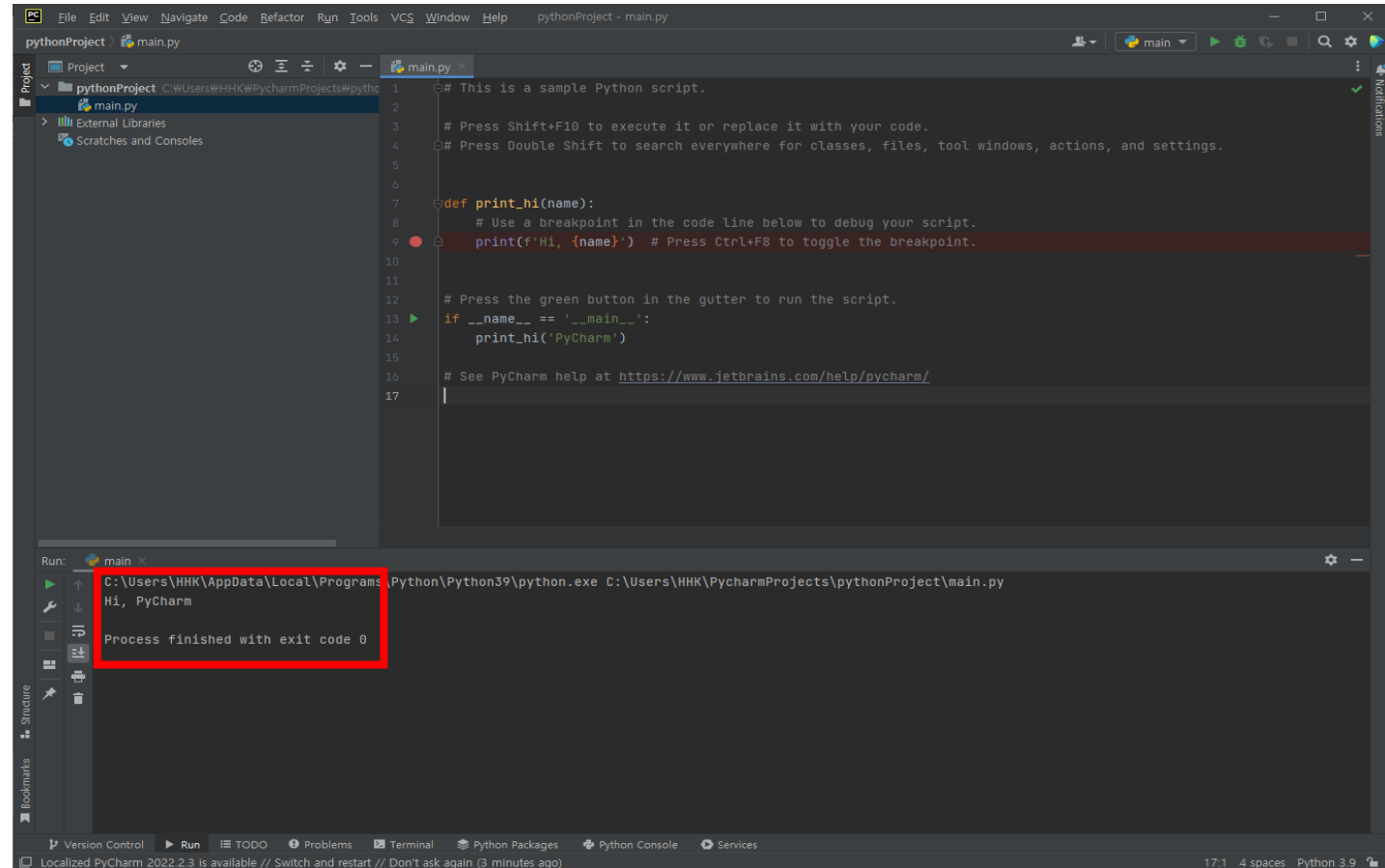
■ Interpreter 설정



PyCharm

■ Interpreter 설정

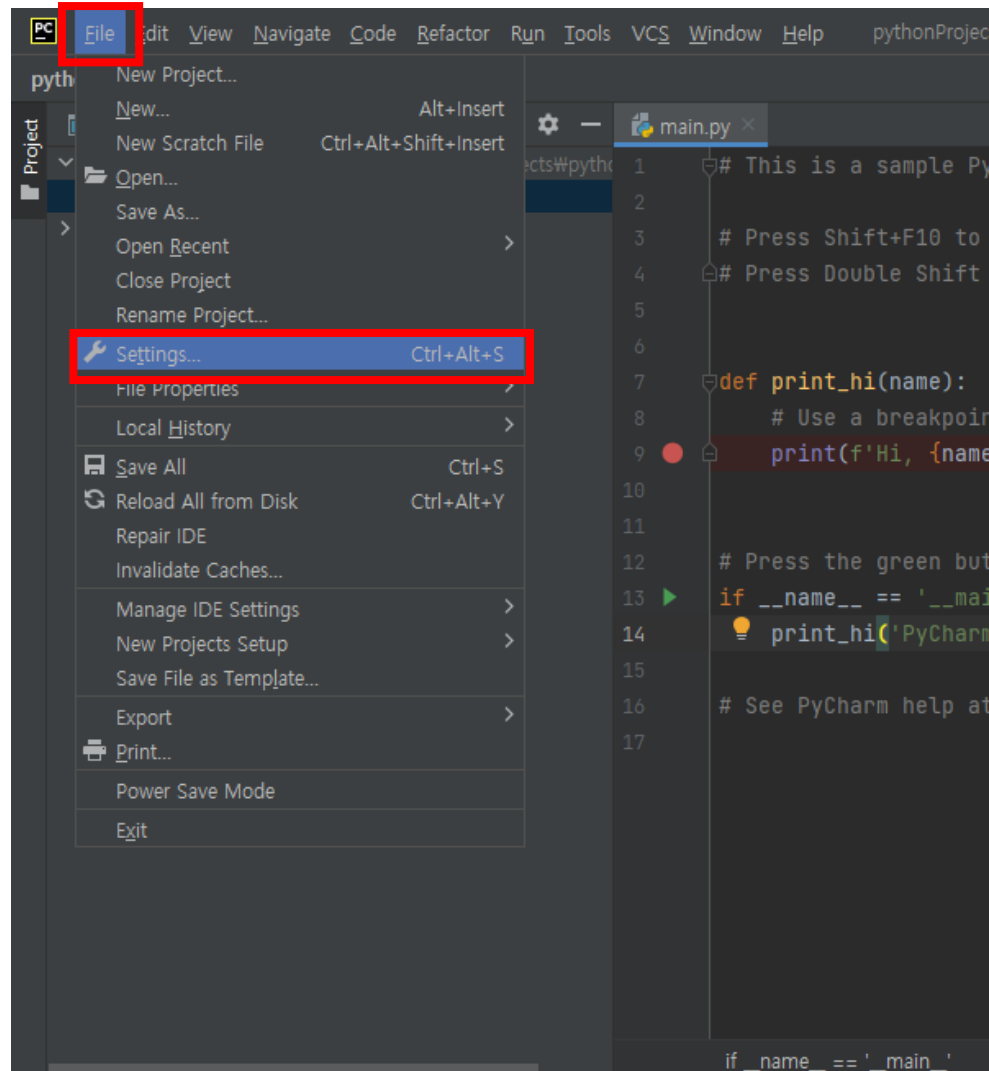
'Hi, PyCharm'
출력 시 성공



Library

■ 라이브러리 설치

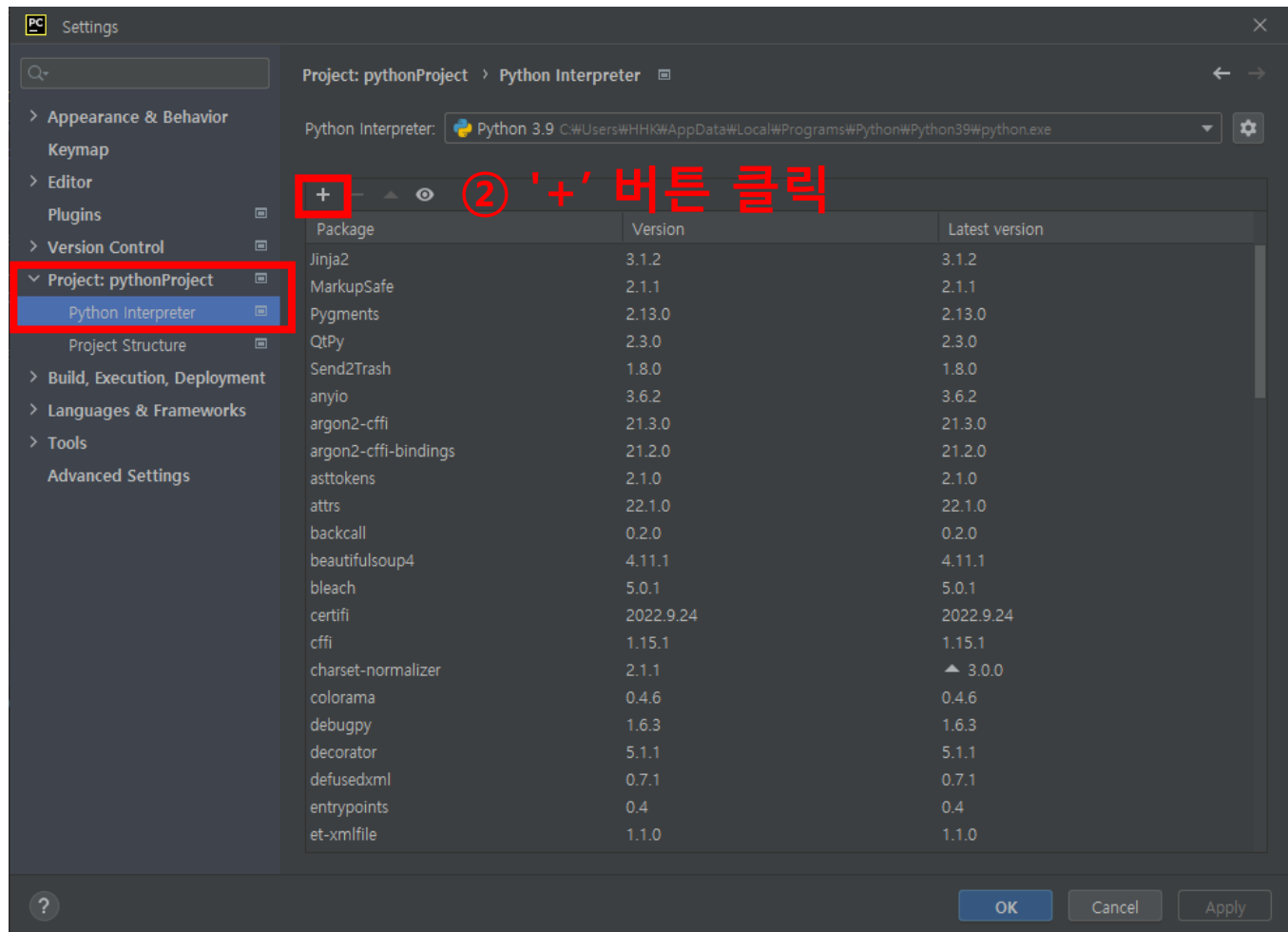
화면 상단
File - Settings 클릭



Library

■ 라이브러리 설치

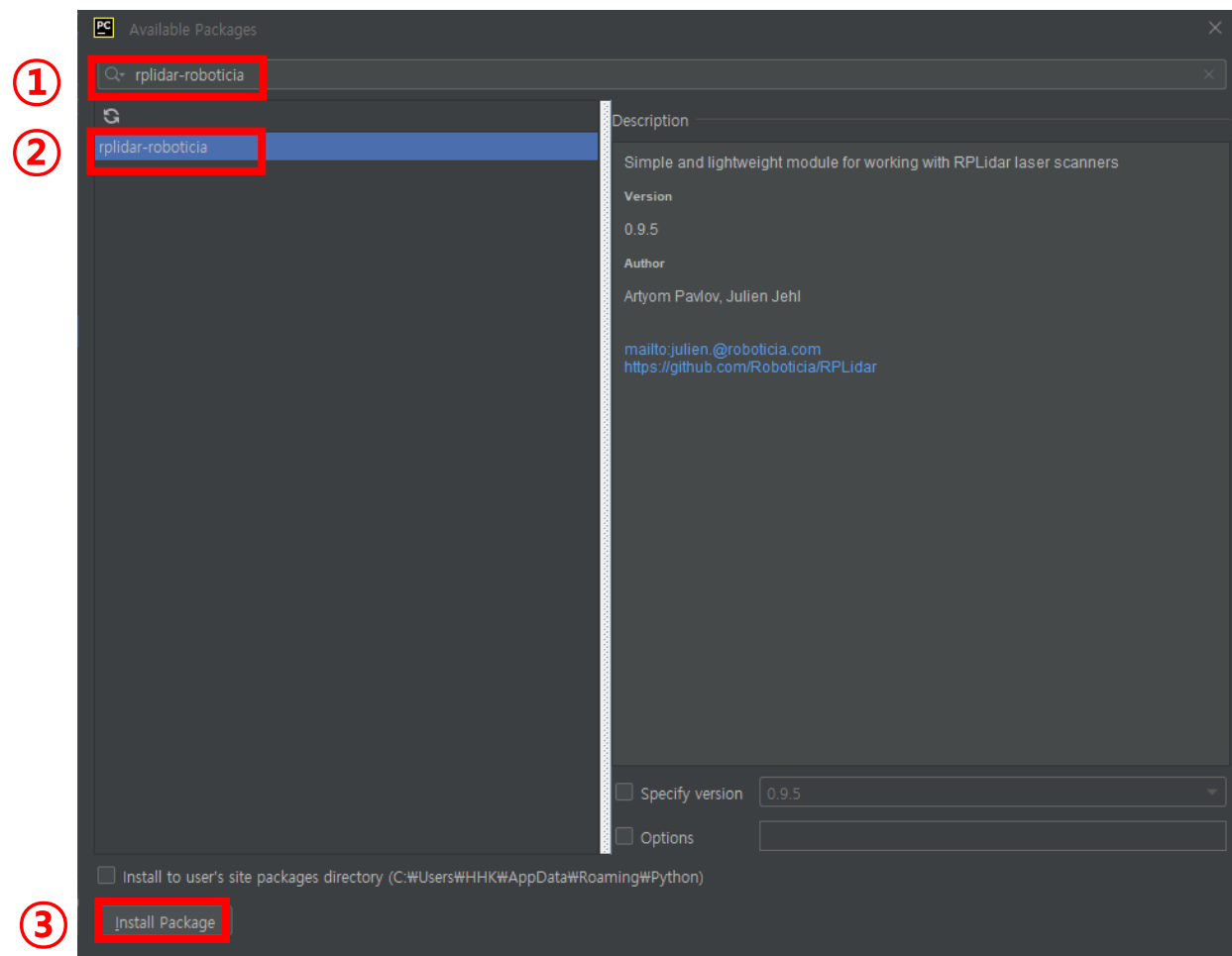
① Project: <프로젝트 이름>
-> Python Interpreter 선택



Library

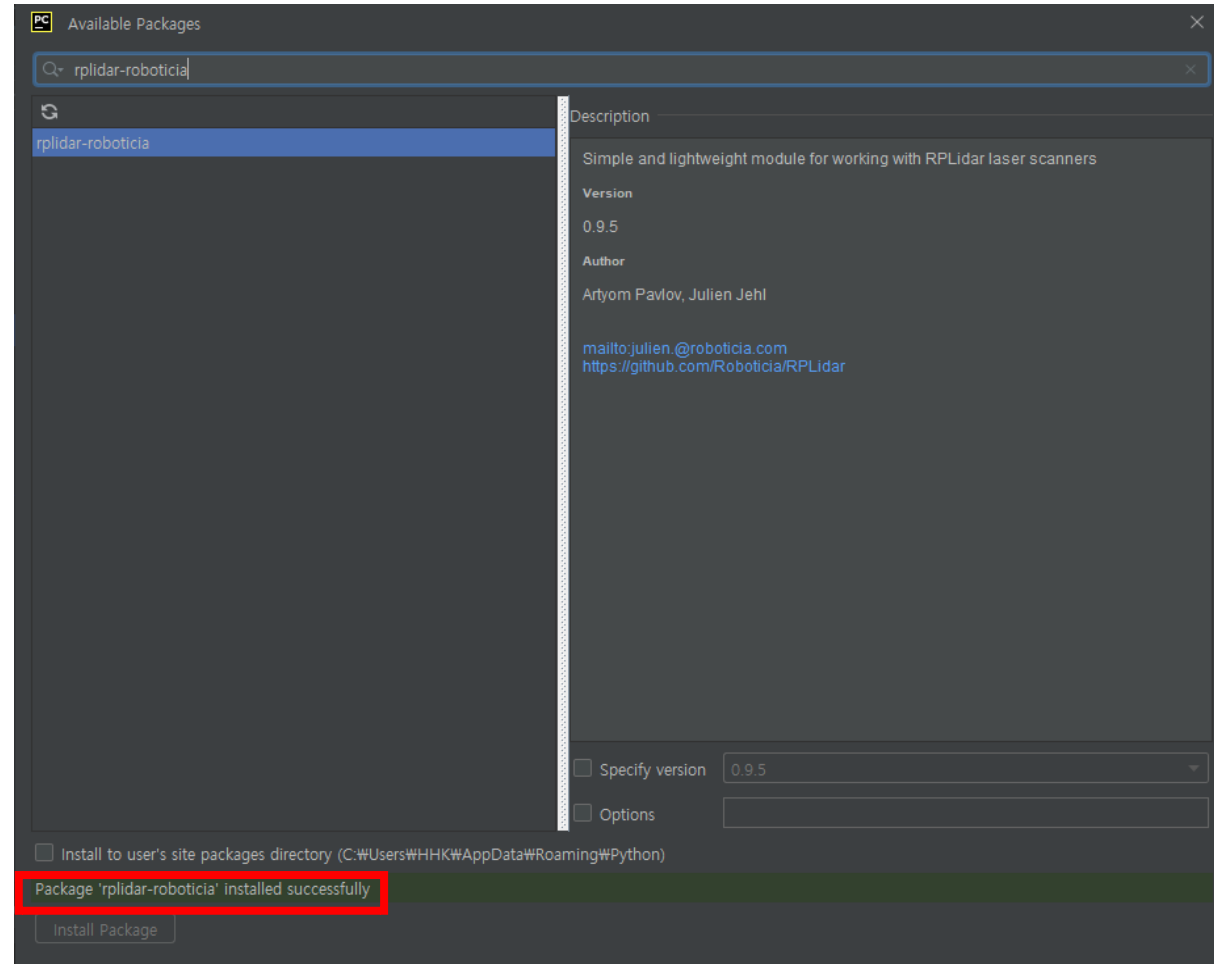
■ LiDAR 라이브러리 설치

- ① rplidar-roboticia 검색
- ② rplidar-roboticia 선택
- ③ Install Package 클릭



Library

■ LiDAR 라이브러리 설치

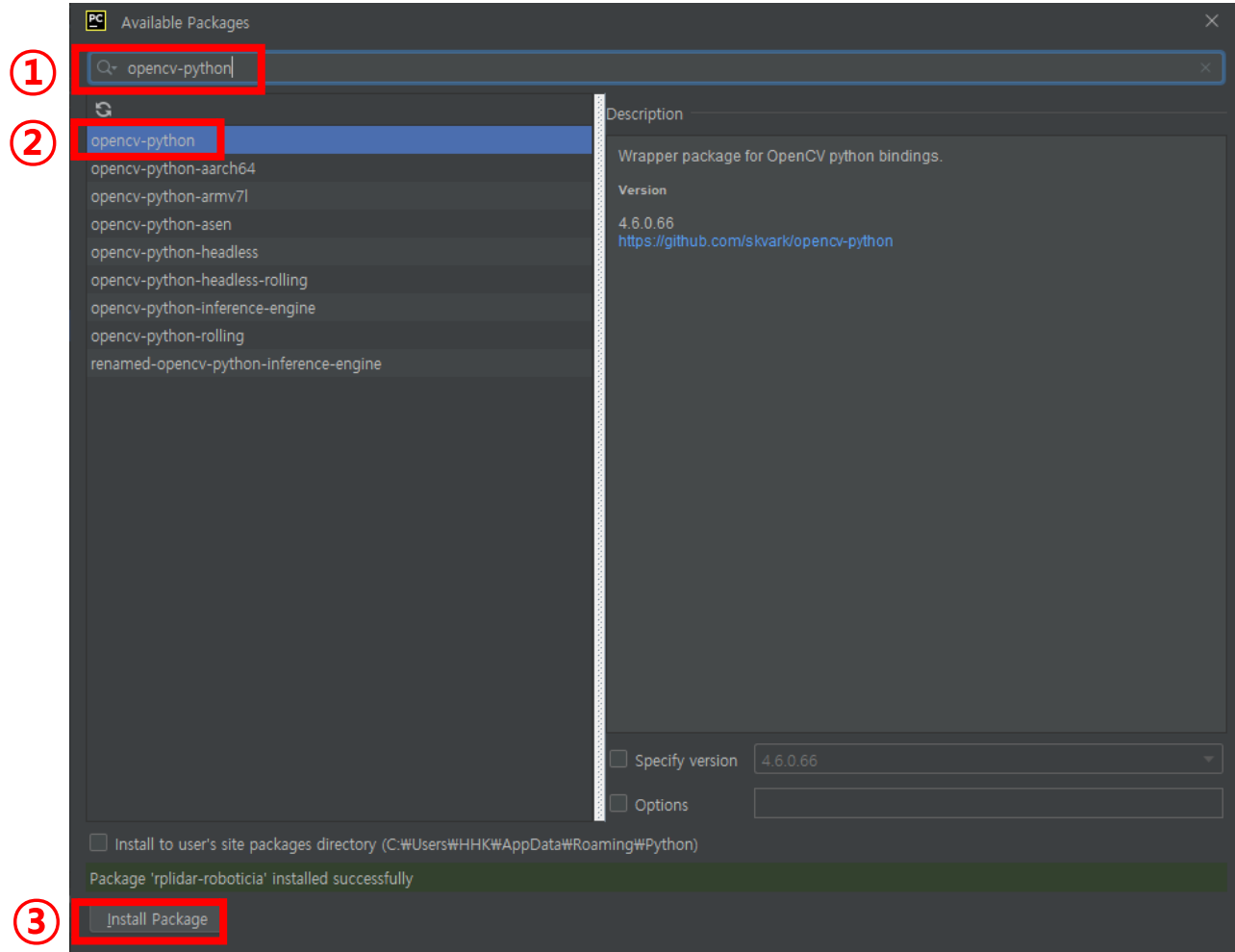


설치 성공 메시지 확인

Library

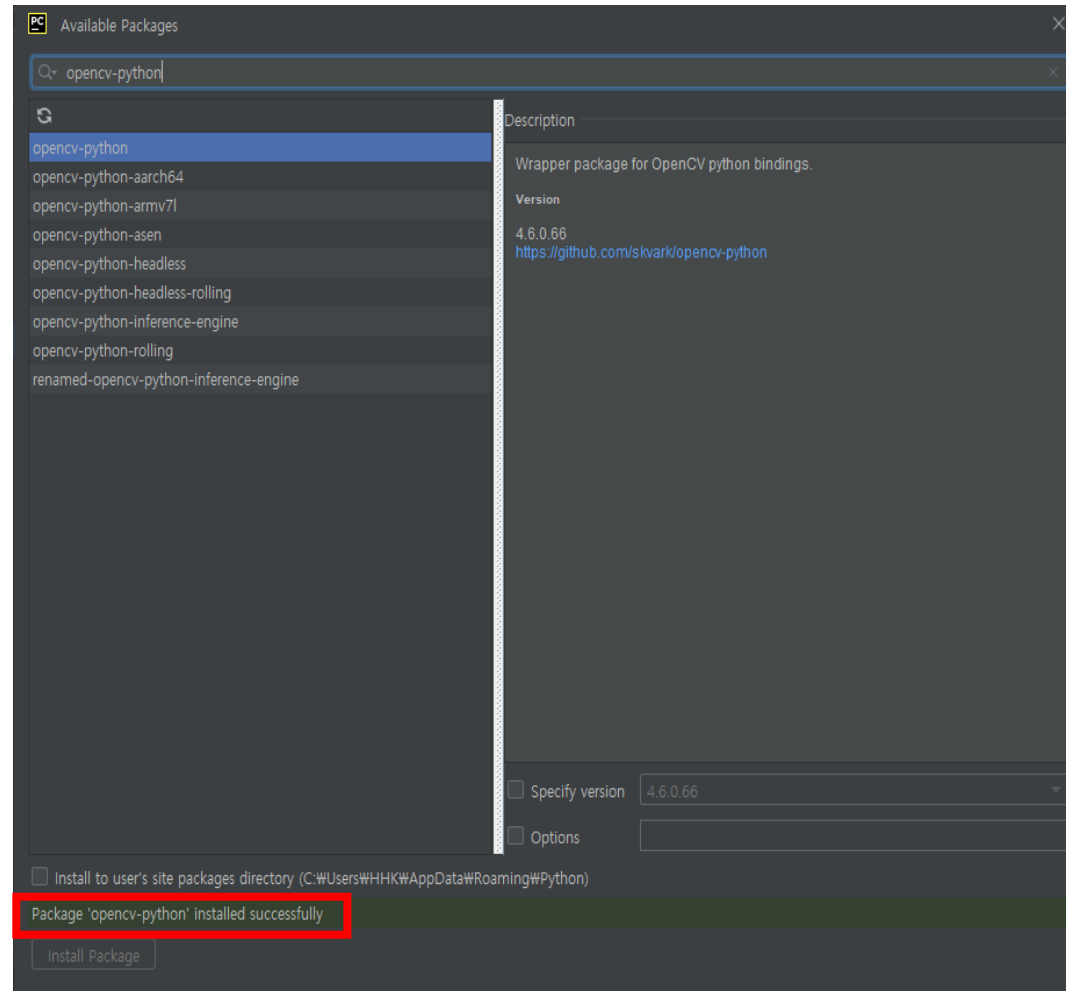
■ Camera 라이브러리 설치

- ① opencv-python 검색
- ② opencv-python 선택
- ③ Install Package 클릭



Library

■ Camera 라이브러리 설치

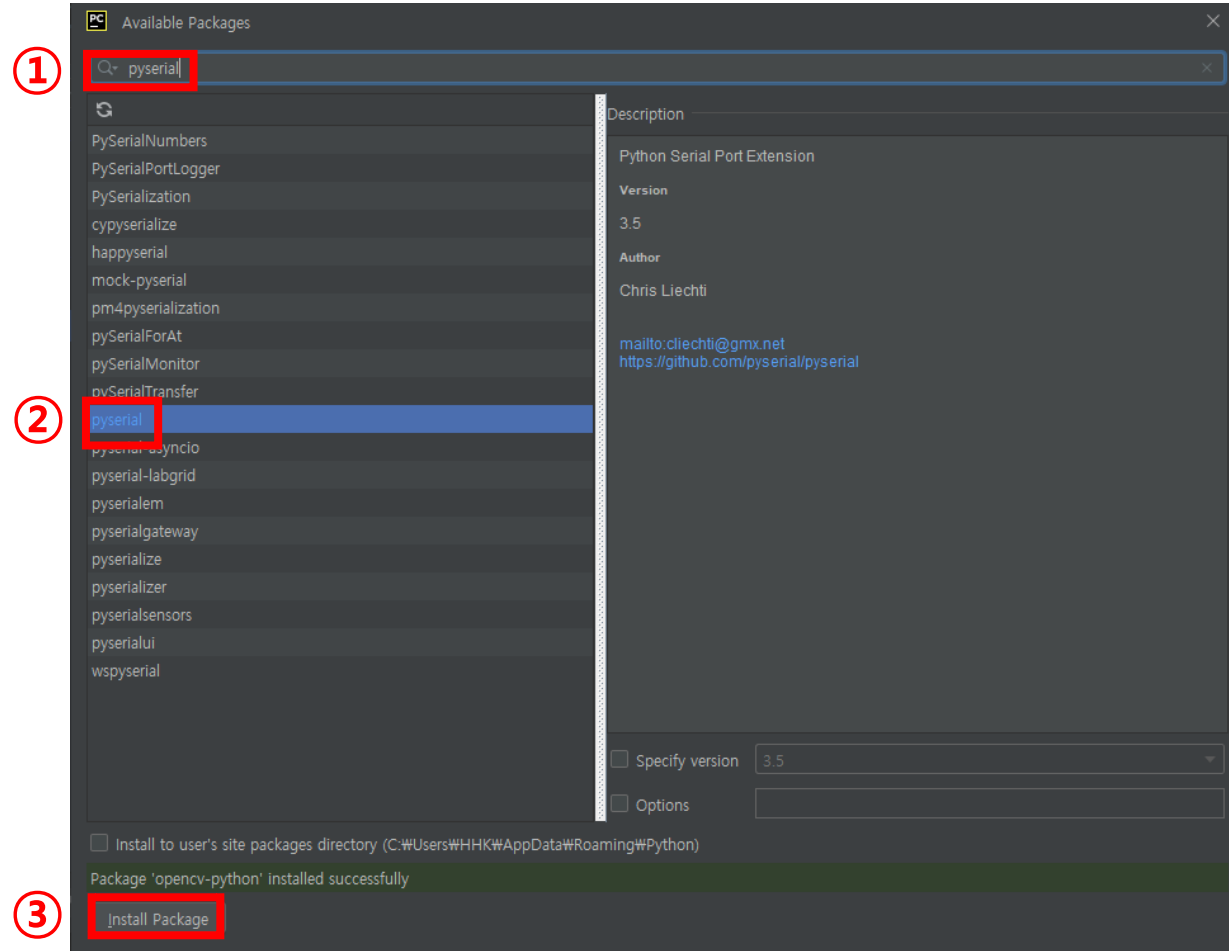


설치 성공 메시지 확인

Library

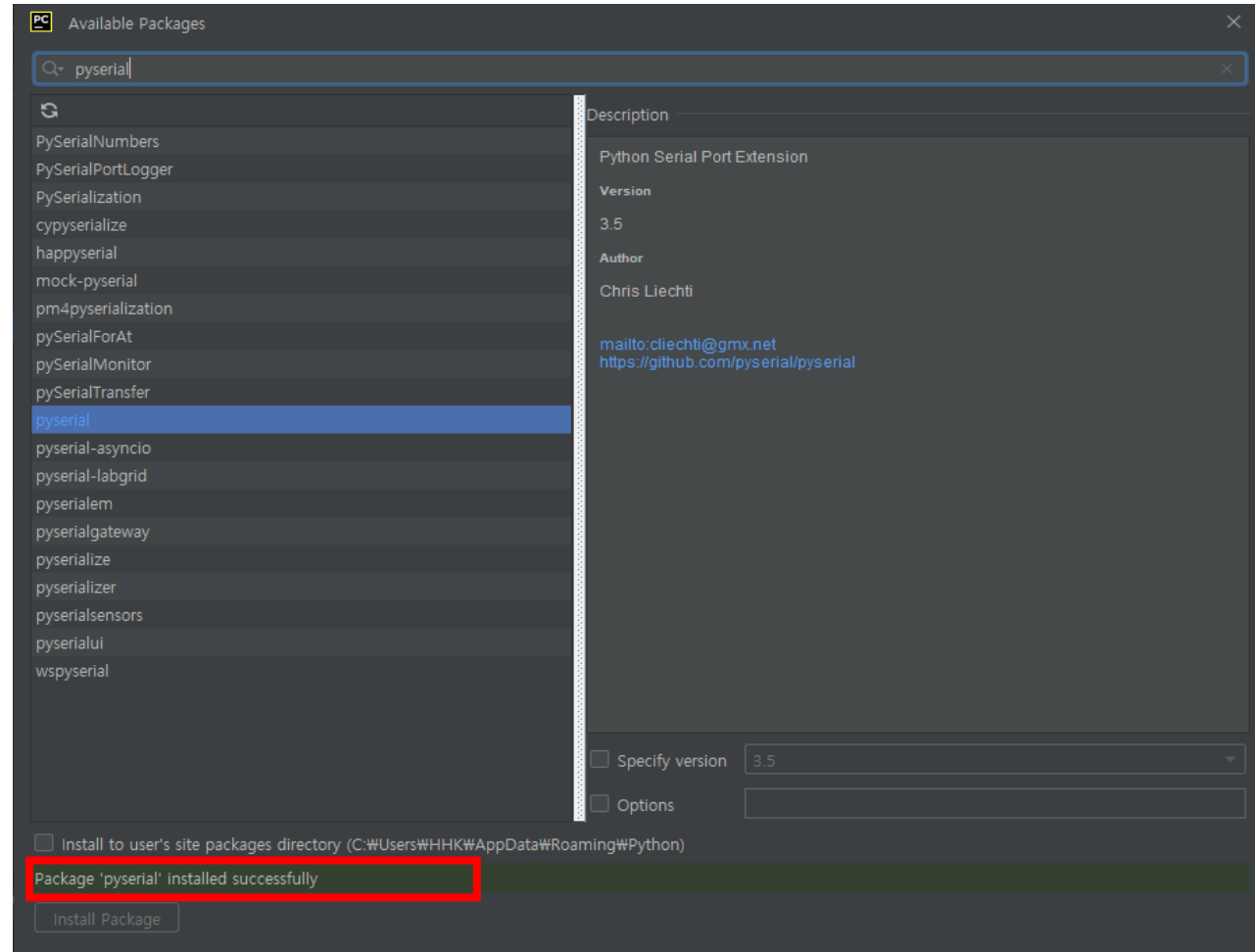
■ 시리얼통신 라이브러리 설치

- ① pyserial 검색
- ② pyserial 선택
- ③ Install Package 클릭



Library

■ 시리얼통신 라이브러리 설치



설치 성공 메시지 확인

Library

■ Matplotlib 설치

① Available Packages

② matplotlib

③ Install Package

matplotlib

matplotlib-arm64

matplotlib-aximgcache

matplotlib-backend-kitty

matplotlib-backend-notcurses

matplotlib-backend-qtquick

matplotlib-backend-wxterm

matplotlib-chord-diagram

matplotlib-colorbar

matplotlib-colors

matplotlib-coolers

matplotlib-dashboard

matplotlib-ephys

matplotlib-extrac

matplotlib-extract

matplotlib-helper

matplotlib-helpers

matplotlib-hep

matplotlib-inline

matplotlib-label-lines

matplotlib-latex-bridge

matplotlib-mercyfan

matplotlib-pdf

matplotlib-pgfutils

matplotlib-polyroi

matplotlib-publishutil

Install to user's site packages directory (C:\Users\YoungsooDo\AppData\Roaming\Python)

Install Package

Description

Python plotting package

Version

3.7.1

Author

John D. Hunter, Michael Droettboom

<mailto:matplotlib-users@python.org>

<https://matplotlib.org>

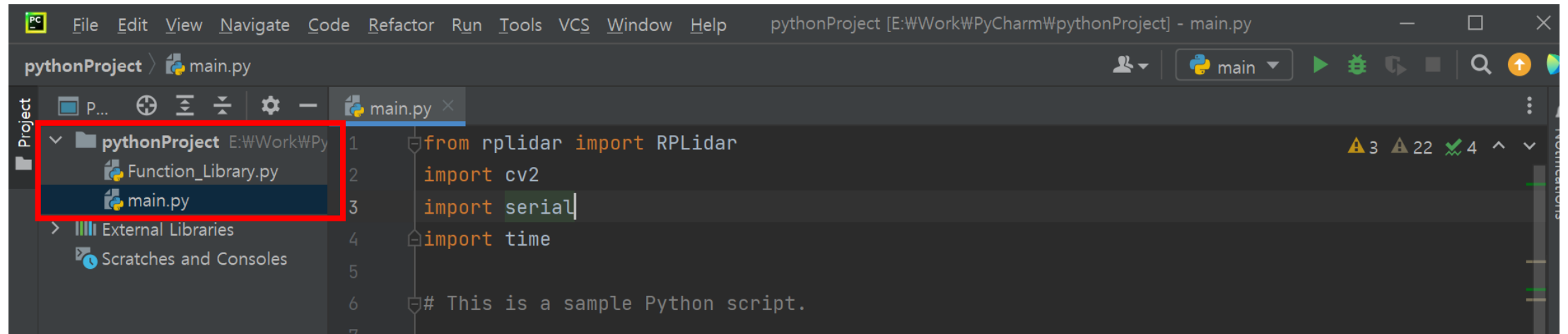
☐ Specify version 3.7.1

☐ Options

Exercise

■ 라이브러리 및 장치 설치 확인

- 새로운 프로젝트를 생성한 후, 기존의 'main.py' 파일 제거
- 배포된 'Function_Library.py' 파일과 'main.py' 파일을 해당 프로젝트로 옮김



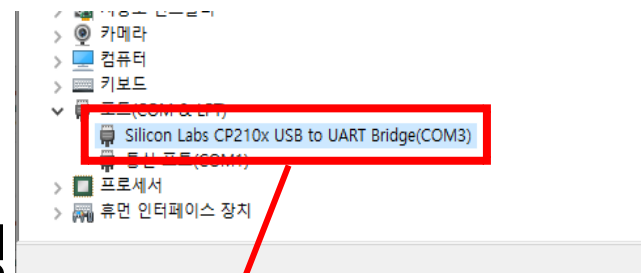
Exercise

장치 관리자 – LiDAR

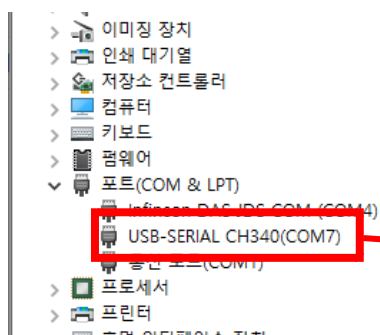
■ 라이브러리 및 장치 설치 확인

● 'main.py' 파일의 init 함수의 parameter 값 변경

- Parm 1 : 아두이노 COM port 번호
- Pram 2 : LiDAR COM port 번호



장치 관리자 – 아두이노



```
Project
└─ pythonProject E:\Work\WPY
   └─ Function_Library.py
      └─ main.py
         └─ External Libraries
            └─ Scratches and Consoles

main.py
56
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else:
    print('\tPort : [X]')

print('\tVersion :', cv2.__version__)

# Press the green button in the gutter to run the script.
if __name__ == '__main__':

    # Parm 1 : Arduino's COM port number
    # Parm 2 : LiDAR's COM port number
    # * the port number must be capitalized.
    # * and we can see this port number on device manager in windows.

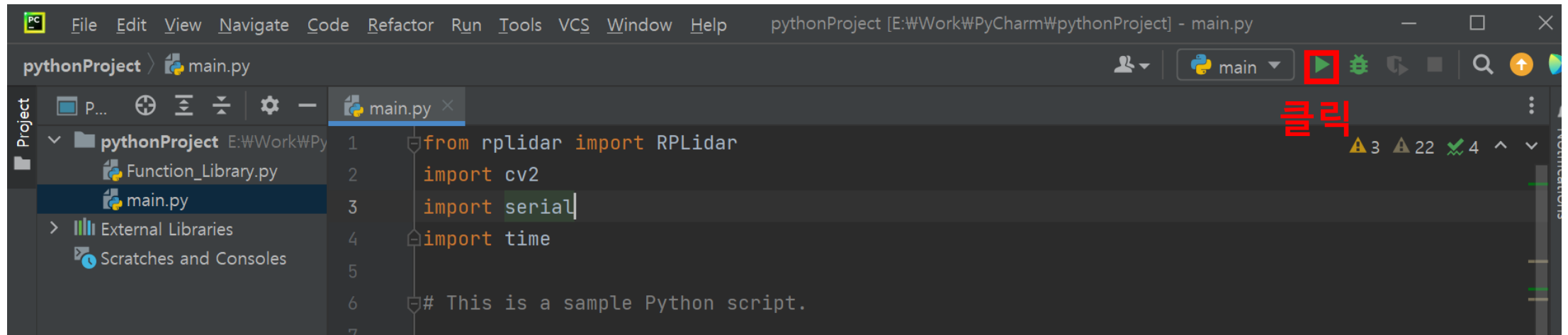
    init('COM5', 'COM11')

# See PyCharm help at https://www.jetbrains.com/help/pycharm/
```

Exercise

■ 라이브러리 및 장치 설치 확인

● Run()



Exercise

■ 라이브러리 및 장치 설치 확인

```
[Arduino] Check
  Lib : [OK]
  Port : [OK]
[LiDAR] Check
  Lib : [OK]
  Port : [OK]
  Info: {'model': 40, 'firmware': (1, 28), 'hardware': 7, 'serialnumber': 'EC8FEDF9C7E29BCEA7E39EF27C5C4304'}
[Computer Vision] Check
  Lib : [OK]
  Port : [OK]
  Version : 4.6.0
```

다음과 같은 문구 출력 시
라이브러리 및 장치 올바르게 설치

Thank You!

Automation Lab.

